Biomagnetostratigraphy of the Vikinghøgda Formation, Svalbard (Arctic Norway), and the geomagnetic polarity timescale for the Lower Triassic

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ABSTRACT

A biomagnetostratigraphy for the Lower Triassic is constructed, using the ammonoid biostratigraphy from Arctic Boreal successions. Combined thermal and alternating field demagnetization determines the Triassic magnetic field polarity in 86% of specimens, with 36% showing linear trajectory line fits and the remainder showing great-circle trends toward the characteristic magnetization. Mean pole directions for the Deltadalen ($\lambda = 50°$, $\varphi = 159°$, dp/dm = 3.9°/5.1°), Lusitaniadalen ($\lambda = 56°$, $\varphi = 163°$, dp/dm = 4.4°/5.4°), and Vendomdalen ($\lambda = 57°$, $\varphi = 143°$, dp/dm = 4.4°/5.4°) members fall close to the European Lower Triassic apparent polar wander path. Mean directions for two of these member-means pass the reversal test. The remanence is predominantly carried by magnetite. The polarity stratigraphy, when integrated with the ammonoid and meager conodont data, is similar to that determined from successions in the Sverdrup Basin (Canada). The Permian-Triassic boundary post-dates a pronounced palynofloral turnover and predates a short duration reverse magnetozone (LT1n.1r). In the correlated Shangsi section (in South China), LT1n.1r occurs after the first appearance datum (FAD) of *H. parvus*, but in the Arctic is within the *Otoceras boreale* Zone. The late Griesbachian to early Smithian is mostly reverse polarity, with three normal-polarity intervals, overlain by mid and late Smithian normal polarity. The Spathian contains four reverse-polarity intervals, the oldest one within the early Spathian with the remainder in the late Spathian. The transition into the Anisian is within the uppermost reverse magnetozone, a feature documented in other sections of this age. The polarity pattern is correlated to other marine sections, indicating the robustness of the biomagnetostratigraphic composite and its utility in calibrating Lower Triassic time.

Keywords: boreal, ammonoids, geochronology, magnetostratigraphy, Permian-Triassic boundary, Svalbard.

INTRODUCTION

Magnetostratigraphy provides a key element in providing high-resolution correlation for the study of the timing of major environmental and climatic changes in the Phanerozoic. For the Triassic there has been much progress made in calibrating the pattern of reverse and normal magnetic field polarity changes against conodont biostratigraphies (Muttoni et al., 1996, 2000, 2004; Gallet et al., 2000; Channell et al., 2003). These studies have predominantly used Tethyan carbonate-dominated marine successions. Successions from the Boreal Triassic in Siberia, the Barents Sea region, and the Sverdrup Basin (Arctic Canada) provide clastic-dominated successions, which are significant for providing ammonoid zonations through the Lower Triassic. The Boreal successions in Arctic Canada are significant for the Lower Triassic in that they have historically provided key type sections for Lower Triassic substages (Tozer, 1967). However, it can be difficult to correlate these in detail to sediment successions deposited in lower latitude (Henderson and Baud, 1997). The highest resolution Lower Triassic ammonoid zonations are those in Siberia (Fig. 1), where some 23 zones and subzonal divisions have been described (Dagys and Weitschat, 1993). This work is a magnetostratigraphic and biostratigraphic study of the Lower Triassic stratotypes on Spitsbergen, comparable to the interval in the Sverdrup Basin studied by Ogg and Steiner (1991). It constructs an ammonoid-calibrated biomagnetostratigraphy for the Lower Trias-

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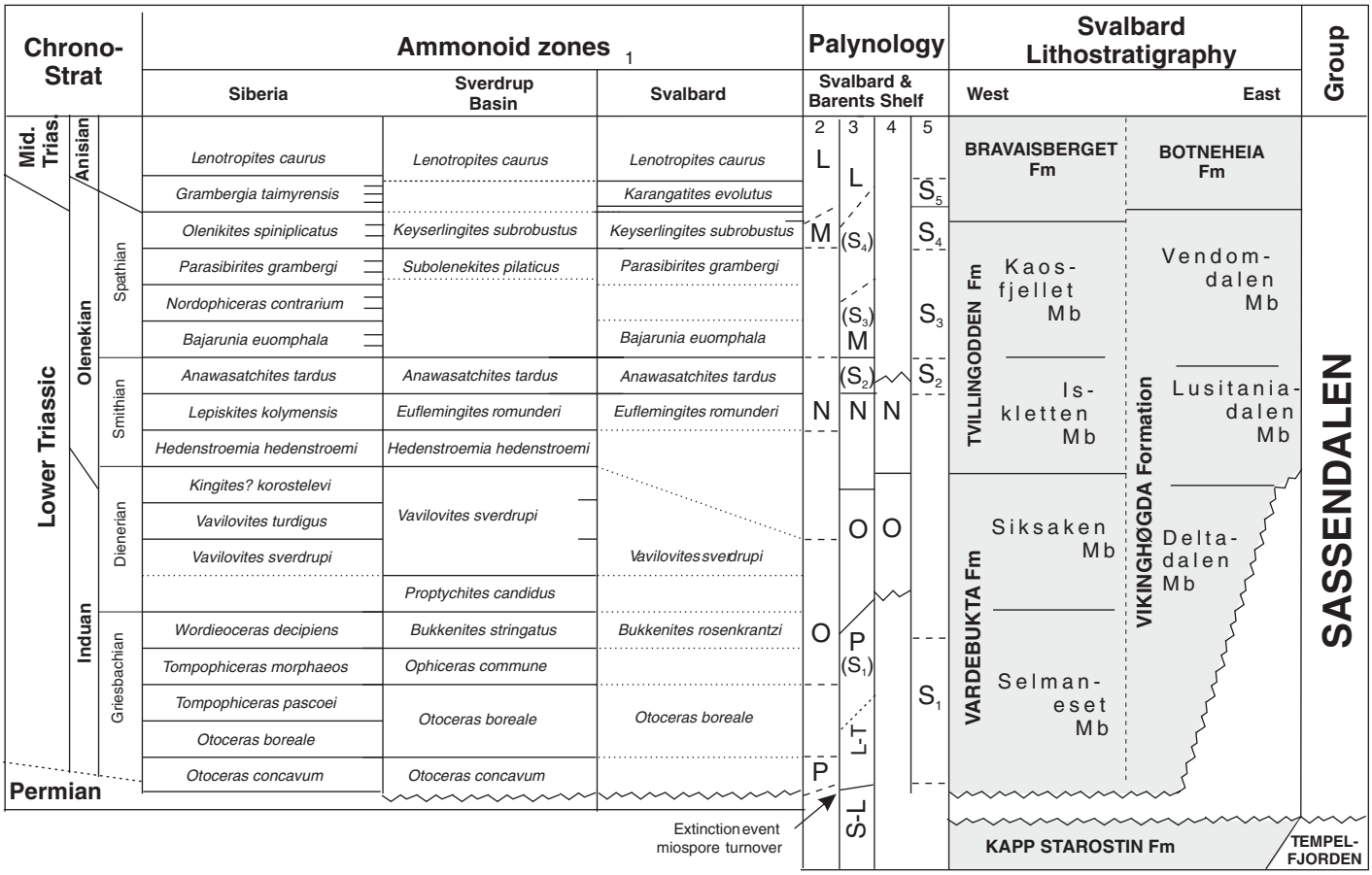


Figure 1. Outline lithostratigraphy and biostratigraphy of Svalbard and some parts of the adjacent Barents Sea. 1—Ammonoid zones (Dagys and Weitschat, 1993; Tozer, 1994; Dagys and Sobolev, 1995). Palynology columns are: 2—Svalbard and Barents Sea region (Hochuli et al., 1989). 3—Vikinghøgda Formation (Fm) (Mørk et al., 1999b, and this work). 4—Bjørnøya (Mørk et al. 1990). 5—Svalvis Dome area of the western Barents Sea (Vigran et al., 1998). S-L—Scutasporites cf. unicus–Lunatisporites spp. Assemblage Zone; L-T—Lundbladispora obsolete–Tympanicysta stoschiana Assemblage Zone. S_1, S_2, etc., are the Svalis$_1$, Svalis$_2$, etc., assemblage zones of Vigran et al. (1998).

sic, which is then used in unifying Boreal and Tethyan biostratigraphies. This in the longer term will lead to improved understanding of the detailed timing of major environmental and biotic changes in the Lower Triassic.

THE SVALBARD LOWER TRIASSIC

The Lower (and Middle Triassic) on Svalbard is essentially polarized into a western and an eastern province, with different lithostratigraphic units (Mørk et al., 1999a). The dominant sediment supply was from the west (Mørk, 1999), resulting in thick deposits of coastal and shallow marine sandstones and shales in western Spitsbergen, whereas marine shales with minor siltstones and sandstones were deposited in basinal settings in central and eastern Spitsbergen, Barentsøya, and Edgeøya (Fig. 1). Within the central and eastern parts of the Svalbard archipelago, the Sassendalen Group is divisible into the Vikinghøgda and Botneheia formations (Mørk et al., 1999b), which span the Lower and Middle Triassic (Figs. 1 and 2).

The Sassendalen Group is well exposed at the mountains Vikinghøgda and Sticky Keep in southern Sassendalen (Fig. 2), where the Vikinghøgda Formation (Fm) is 250 m thick, and is divided into three members (Fig. 1; Mørk et al., 1999b). The lowest, the Deltadalen Member (Mb) is exposed in the river banks in Deltadalen (the sampled section DR; Fig. 2), the valley between Vikinghøgda and Sticky Keep. The Lusitaniadalen Mb is exposed in the riverbanks of the tributary continuing around the southern foot of Vikinghøgda (VR section; Fig. 2). The lower part of the Vendomdalen Mb is exposed in the same tributary and riverbanks, while the upper part of this member is best exposed in the surrounding mountain cliffs, but was measured and sampled on the E-facing side of Vikinghøgda (section VM; Fig. 2). The Vendomdalen Mb succession in these two sections can be correlated by following a pronounced bivalve-rich dolomite bed at a level 173 m above the base of the formation. These three sections combined give an almost continuous exposure from the Permian Kapp Starostin Fm, through the Vikinghøgda Fm, to the Middle Triassic Botneheia Fm.

The basal part of the Deltadalen Mb is sandstone resting on an erosional surface of partly silicified siltstones of the Kapp Starostin Fm (Fig. 3; GSA Data Repository Fig. DR1[1]; and Figs. 6A–6C in Mørk et al., 1999b). Above the basal sandstone, the Deltadalen Mb consists of silty shale with intervals dominated by siltstone or very fine-grained sandstones, often with hummocky cross-lamination. There is evidence of two coarsening-up units (Fig. 3), the upper one

[1]GSA Data Repository Item 2008106, the nature of the member boundary intervals and details on the nature of the overprint magnetization, is available at www.geosociety.org/pubs/ft2008.htm. Requests may also be sent to editing@geosociety.org.

terminating the formation boundary. The transition into the base of the Lusitaniadalen Mb is marked by a progressive reduction in the number of thin sandstones within the silty-mudstone succession, without evidence of erosional contact (Fig. DR1; see footnote 1).

The base of the Lusitaniadalen Mb is an interval of dark-gray laminated silty mudstone with a few thin sandstone or siltstone beds (Fig. DR1 [see footnote 1] and Figs. 6F and 6G in Mørk et al., 1999b). The parallel-laminated sandstone and siltstone beds are a prominent feature of the lower ~55 m of this member. Small (20–30 cm in diameter), globular or flattened carbonate nodules occur in a few horizons from ~15 m above the base of the member, and a prominent horizon of large septarian concretions is found 25 m above the base (Fig. 3). Calcite concretions are abundant in the middle and upper parts of the member. These concretions formed early in diagenesis, evident by deformation of the enclosing shales, and the uncrushed nature of the enclosed fossils. The uppermost part of the Lusitaniadalen Mb is an ~0.9 m thick ferroan dolomite cemented silty-sandstone (locally cross-bedded) with interbeds of gray mudstone. This sandstone is similar in character to sandstone beds lower in the Lusitaniadalen Mb, although without the dolomite cement.

The Vendomdalen Mb consists of silty, dark-gray, laminated mudstones, thin- to medium-bedded, with late diagenetic, yellow-weathering, ferroan dolomitic beds and nodules, a distinctive feature of the member throughout central Spitsbergen. The basal ~10 cm of the Vendomdalen Mb shows a transition from the uppermost bed of the Lusitaniadalen Mb with an increase in shale content into the basal part of the Vendomdalen Mb. There is no evidence of erosion at this boundary (Fig. DR1; see footnote 1). The basal few meters of the Vendomdalen Mb consist of dark-gray shale (with infrequent calcite concretions), which is significantly darker colored than that above ~160 m, and lacks the yellow-weathering dolomitic concretions common throughout most of the member. Above the silty uppermost bed at 250 m is a very dark-gray shale with small phosphate nodules indicating the base of the Botneheia Fm (Fig. 3). There is no apparent sedimentological evidence of erosion at this contact. Seven meters above the base of the Botneheia Fm a dolomitic siltstone bed, bioturbated by *Rhizocorallium* (at 257 m), is a local marker in the Botneheia Fm of the Sassendalen area.

The hiatus at the base of the Deltadalen Member is part of a regional disconformity described from other areas of Svalbard, where sediments as young as Dienerian may rest on the Kapp Starostin Fm (Mørk et al. 1989). The boundar-

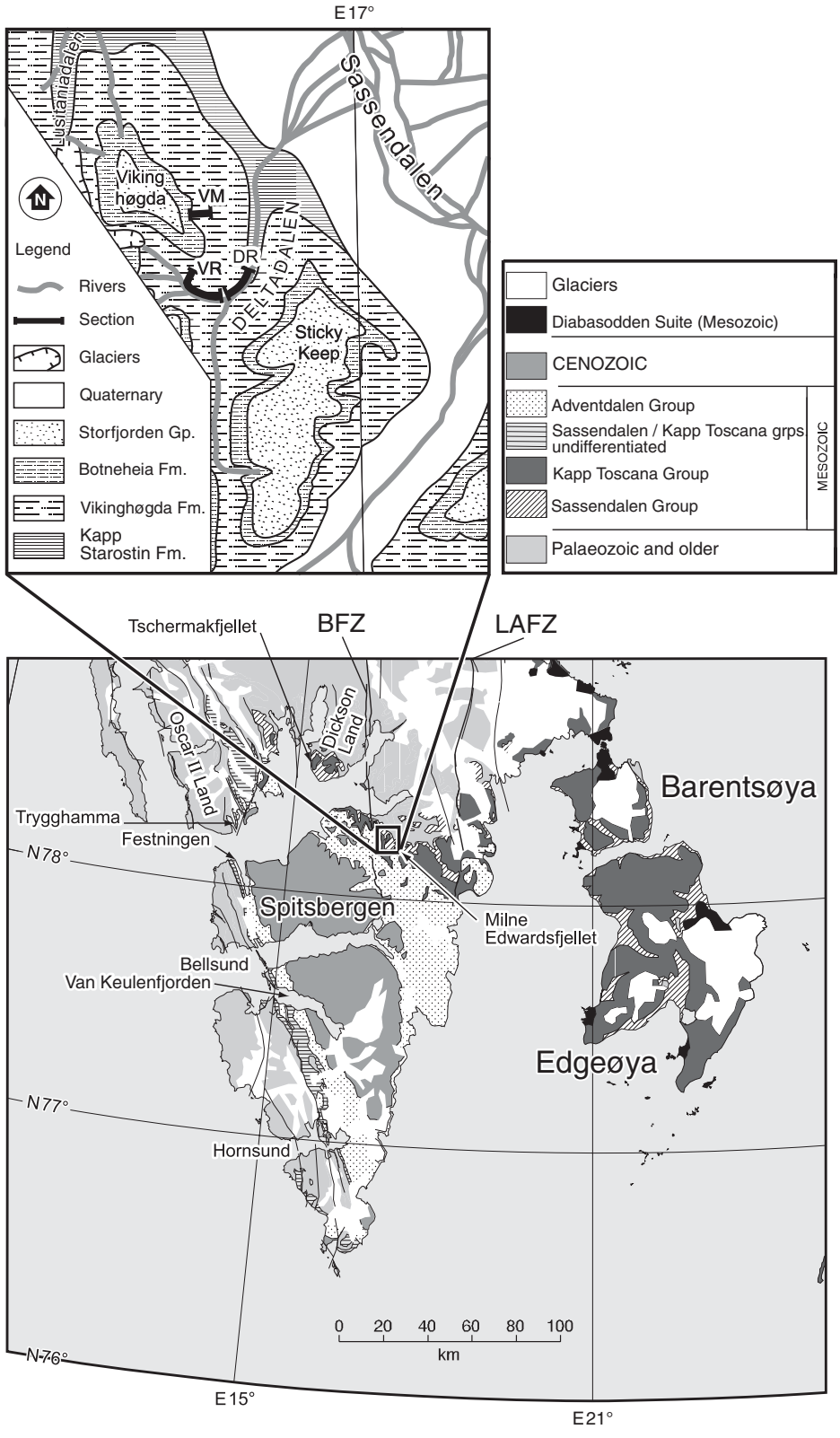


Figure 2. Location map of Deltadalen in central Spitsbergen and a simplified geology of southern Svalbard (modified from Dallmann, 1999). Separate legends for main map and inset. BFZ—Billefjorden Fault Zone; LAFZ—Lomfjorden-Argardbukta Fault Zone. Inset shows the location of the sections in Deltadalen (DR, VR, and VM) detailed here.

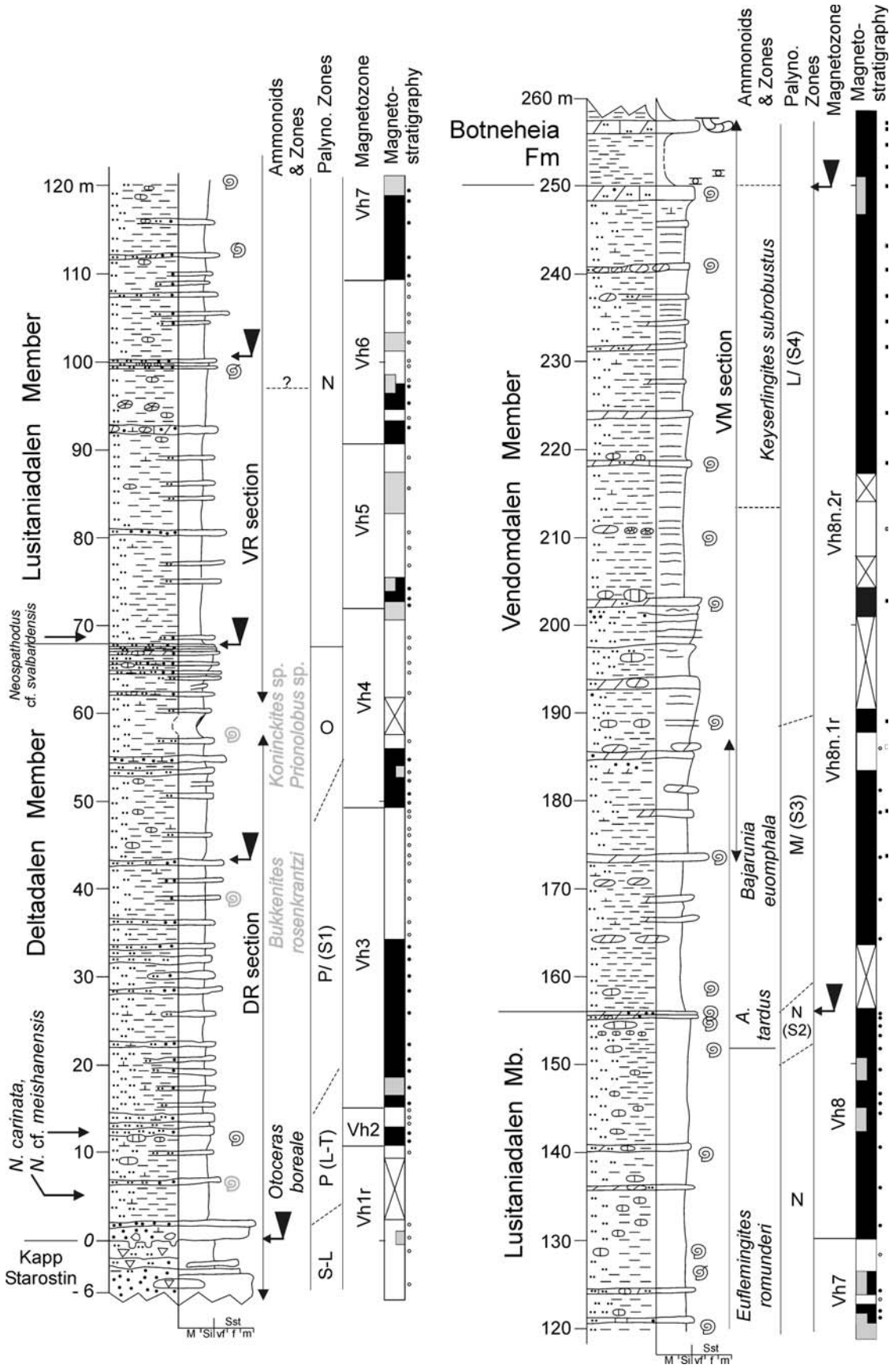


Figure 3. Summary lithological log (Mørk et al., 1999b), magnetostratigraphy, and significant biostratigraphy of the Vikinghøgda Formation (Fm) in the Deltadalen area. Ammonoids data in gray, found in correlated section (see text). Magnetostratigraphy: black—normal; white—reverse; gray—uncertain; half-bar width—single specimen of less than best quality paleomagnetic data. X—magnetostratigraphic sampling gap. Palynological assemblage zones as in Figure 1.

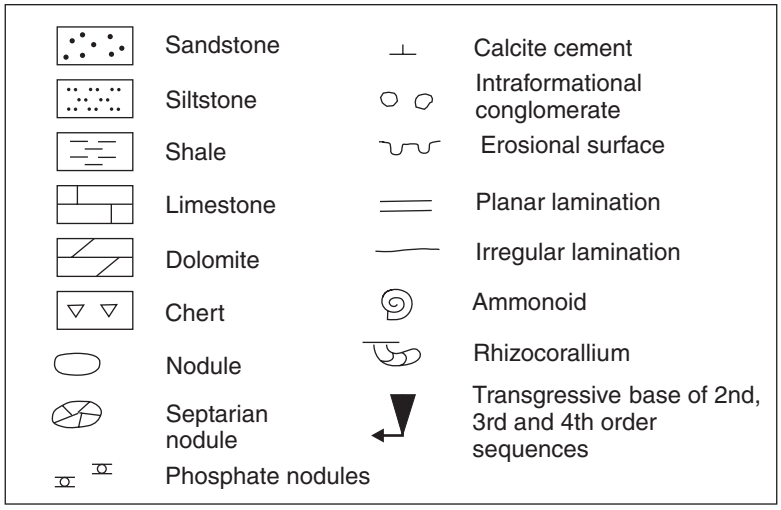


Figure 3 (*continued*).

ies at the base and tops of the other members at Vikinghøgda are conformable in character without evidence of erosional contact. In areas of western Spitsbergen, equivalent boundary intervals also appear to be conformable (i.e., in the Vardebukta and Tvillingodden formations; Fig. 1) but show a greater dominance of sandstone in the regressive (low-stand) part of the units, due to closer proximity to the basin margin (Mørk et al. 1982). The basal few meters of the Deltadalen, Lusitaniadalen, and Vendomdalen members at Vikinghøgda (in case of the Deltadalen Mb, above the basal sandstone) is a shale or mudstone without substantive sandstone or siltstone beds, which generally grades upward into mudstone with thin sandstone or siltstone beds. These lithological changes are interpreted as transgressive units at the base of each member, and mark the backstepping of sources of sandstone and siltstone from the Lower Triassic basin, a response which can be mapped throughout Svalbard (Mørk et al., 1989).

BIOSTRATIGRAPHY

The ammonoid, conodont, and miospore biostratigraphy in the sections has been reported by Mørk et al. (1999b), with ammonoids providing the most crucial age control. In common with many other sections in the Boreal Triassic, macrofossil occurrence is often sporadic—as a result of preservation often only in concretions (Weitschat and Dagys, 1989). Only miospore data provide relatively continuous stratigraphic preservation and are also significant in that they allow a wider regional zonation based on assemblage zones defined from Svalbard, the southern Barents Sea (Hochuli et al., 1989; Mangerud, 1994), and the Svalis Dome area (400 km to south) of the western Barents Sea (Vigran et al., 1998).

Late Permian and Induan

Conodonts from 5 m below the figured section (Fig. 3) and at 0.5 m show fragments of *Mesogondelella*, which are not age diagnostic, since they are common throughout the Kapp Starostin Fm (Nakrem et al., 1992; Mørk et al., 1999b). Most biostratigraphy suggests that the Changhsingian and the Wuchiapingian (most of Dzhulfian) are absent from the Kapp Starostin Fm (Nakrem et al., 1992; Wignall et al., 1998), although on the basis of the $\delta^{13}C_{org}$ record in the Festningen section, in western Spitsbergen (Fig. 1), it is clear the uppermost Changhsingian is present in the basal few meters of the Vardebukta Fm (Wignall et al., 1998). Miospores from the Kapp Starostin Fm and basal 1.7 m of the Deltadalen Mb indicates the equivalent of the Scutasporites unicus–Lunatisporites Assemblage Zone, described from the uppermost part of the Tempelfjorden Group on the Finnmark Platform (Mangerud, 1994), indicating a possible Upper Permian age.

A single macrofossil-bearing horizon occurs in the studied section in the Deltadalen Mb. Carbonate concretions, 11.5 m above the base, contain a fauna of ammonoids *Otoceras boreale, Tompophiceras* cf. *gracile* along with bivalves *Claraia stachei* and *Adontophora* sp. The co-occurrence of *O. boreale* and ophiceratids indicates that this level represents the upper part of the *Otoceras boreale* Zone (Weitschat and Dagys, 1989; Dagys and Weitschat, 1993). A Boreale Zone fauna has also been described in the very similar succession from Lusitaniadalen 4 km to the NW (Mørk et al., 1999b; Fig. 2), where a fauna of *O. boreale* with *Glyptophiceras nielseni* and other ophiceratids is found in two horizons 6 m and 9–10 m above the base of the Deltadalen Mb (Korchinskaya and Vavilov, 1987).

From the upper part of the Spitsbergen Griesbachian *Bukkenites rosenkrantzi* is known from Lusitaniadalen and *B.* cf. *strigatus* from Dickson Land (Korchinskaya, 1986; Korchinskaya and Vavilov, 1987). The occurrence of *B. rosenkrantzi* in the Lusitaniadalen section is some 25 m above the uppermost *Otoceras boreale* horizon, which equates to a level in the Deltadalen section at ~37 m (Fig. 3). Some 15–20 m above this level (equivalent to 52–57 m at Deltadalen), *Koninckites* sp. and *Prionolobus* sp. occur. *Koninckites* characterizes the oldest subzone of the Sverdrupi Zone (late Dienerian) of Tozer (1994), although in China this genus ranges up into the Olenekian.

The conodont *Neogondolella (= Clarkina) carinata* has been recovered at the 5.0 m and 11.5 m levels (Mørk et al., 1999b), and is commonly found in association with *Otoceras* on Spitsbergen (Dagis and Korchinskaya, 1987). This species is well known from the lowermost Triassic sections in western and Arctic Canada (Henderson, 1997; Henderson and Baud, 1997), where it ranges from the Boreale Zone to the Strigatus Zone (Orchard and Tozer, 1997). These conodont assemblages in the Deltadalen Mb also contain *Ng. meishanensis* (Nakrem et al., 2006), a species which is typical of the latest Permian at Meishan (Yin et al. 2001), but ranges into beds containing *Hindeodus parvus* in Arctic Canada (Henderson and Baud, 1997).

The miospore assemblage in the section between 4.7–14.0 m is correlated to the Lundbladispora obsoleta–Tympanicysta stoschiana Assemblage Zone (Mangerud, 1994) and the closely equivalent Assemblage P of Hochuli et al. (1989). At Vikinghøgda, this assemblage contains forms such as *Vittatina*, *Lueckisporites virkkiae,* and *Inaperturopollenites nebulosus,* which are also typically found in late Permian strata, along with miospores appearing in the lowest Triassic such as *Densoisporites* sp., *Rewianispora* sp., *Lundbladispora* sp., and *Kraeuselisporites* sp. (Mørk et al., 1999b). The mixed Permian-Triassic character of this assemblage is equivalent to similar transitional microflora, following the late Permian extinction level, commonly seen in Permian-Triassic transitional strata (Shu and Utting, 1990; Utting, 1994; Looy et al., 2001; Yaroshenko and Lozovsky, 2004; Yaroshenko, 2005). Hence the late Changhsingian extinction level is located between 1.7 and 4.7 m in the section (Fig. 3).

The equivalent of the $Svalis_1$ assemblage (20.3–48.0 m in the section) co-occurs with ammonoids *Axelia* sp. aff. *elegans* and *Ophiceras* sp., in the Svalis Dome area of the Barents Sea, indicating the late Griesbachian Commune Zone (Vigran et al., 1998). Equivalent assemblages to those between 4.7 and 48.0 m in the

section occur in the Concavum, Boreale, and Commune Zones (and equivalents) in Canada and Greenland (Balme, 1980; Piasecki, 1984; Utting, 1994).

A single element of the conodont *Neospathodus* cf. s*valbardensis* occurs at 69 m (Mørk et al., 1999b). At Diener Creek in Artic Canada this species occurs in the Dienerian Candidus Zone in association with the conodont *Ns. dieneri,* which commonly ranges throughout the Dienerian in Canada (Orchard and Tozer, 1997). The late Dienerian ammonoid *Vavilovites* aff. *sverdrupi* has been recorded from the Bellsund area (Gazdzicki and Trammer, 1977; Korchinskaya and Vavilov, 1987; Fig. 2), Lusitaniadalen, the Van Keulenfjorden region (Korchinskaya, 1986; Weitschat and Dagys, 1989), and at Bertilryggen (Oscar II Land).

Olenekian

The higher part of the Lusitaniadalen Mb is the only unit within the Svalbard Triassic in which it is possible to separate faunal zones based on the height ranges of ammonoids. The lowest fossiliferous level in the section (98 m, Fig. 3) contains only large arctoceratids that are typical of the Romunderi Zone in the Sverdrup Basin (Tozer, 1994) and in Spitsbergen. The index ammonoid *Euflemingites romunderi* has its first occurrence at 128 m (Fig. 3).

Ammonoids from the Romunderi Zone in the section (apart from the index species) are *Arctoceras blomstrandi*, *Paranannites spathi,* and *Anakashmirites* sp. The bivalve *Posidonia mimer* is also characteristic of this zone and occurs in the type section at several levels (Mørk et al., 1999b). In the sections, the highest ammonoids from the Romunderi Zone occur at 152 m (Fig. 3). Ammonoids from the carbonate nodule layer at 154.5 m (Fig. 3) indicate the late Smithian Tardus Zone, which extends upward from this level. This zone is widely encountered in all sections in central Spitsbergen (Dickson Land, Oscar II Land, and Sassendalen area), the east coast of Spitsbergen, Barentsøya, and Edgeøya. The bivalve *Pseudomonotis occidentalis* also characterizes this level and is known from contemporaneous beds in Artic Canada. In the boundary beds of the Lusitaniadalen-Vendomdalen members, *Xenoceltites subevolutus* is present, indicating the Tardus Zone and showing the transition to the Vendomdalen Mb takes place within the Tardus Zone.

In the uppermost levels of the Lusitaniadalen Mb, a relative increase in the abundance of miospores *Densoisporites nejburgii*, *Cycadopites* sp., and fungal remains is inferred to indicate similarities to the $Svalis_2$ assemblage (Mørk et al., 1999b; Vigran et al., 1998). In the Svalis dome area, this assemblage co-occurs with *Arctoprionites nodosus* and the bivalves *Posidonia mimer* and *Pseudomonotis occidentalis* (Vigran et al., 1998), consistent with the Tardus Zone fauna in the section. The restricted thickness of the $Svalis_2$ assemblage in the section, compared to greater relative thicknesses in the Barents Sea (Vigran et al., 1998), suggests some condensation (or omission) at about this level, a feature also known from some areas of the Barents Sea (Van Veen et al., 1992). Evidence of erosion at this level is absent from the section, although the regressive (low-stand) character of the uppermost Lusitaniadalen Mb is clear by the presence of the sandstone at this level (Fig. 3 and Fig. DR1 [footnote 1]).

Fossils in the Vendomdalen Mb dolomitic beds and concretions are mostly badly preserved and usually compressed. At 173 m, ammonoid imprints belong to *Bajarunia* ex gr. *euomphala* indicating the Euomphala Zone of the lowermost Spathian. In the Vendomdalen Mb at Milne Edwardsfjellet (16 km to the southeast), the ammonoid *Parasibirites* cf. *elegans* indicates the Grambergi Zone of the late Spathian (Figs. 1 and 3). In the Svalis Dome area of the Barents Sea, the $Svalis_3$ assemblage (equivalent to 159–188 m in the section) co-occurs with the ammonoid *Pseudosageceras*, which has an age range from the Tardus Zone to the late Spathian (Vigran et al., 1998).

The ammonoid *Keyserlingites* sp. occurs from the ~214 m to 241 m level, a genus which is restricted to the late Spathian (Dagys and Weitschat, 1993). A carbonate nodule layer at 241 m (Fig. 3) contains a well-preserved latest Spathian fauna, which includes *Keyserlingites subrobustus*, *Svalbardiceras spitzbergense,* and *Popovites occidentalis,* and the bivalve *Posidonia aranea*, which is restricted to the Subrobustus Zone of the Boreal Triassic (Weitschat and Dagys, 1989). *Svalbardiceras spitzbergense* indicates the youngest subzone of the Siberian Olenekian (Dagys and Sobolev, 1995). No macro fauna were found in the Botheheia Fm in the VM section.

PALEOMAGNETIC METHODS AND ANALYSIS

In total, 289 paleomagnetic specimens were collected through the succession, mostly from siltstone and sandstone beds in the Deltadalen and lower parts of the Lusitaniadalen members, from concretionary calcite nodules in the upper part of the Lusitaniadalen Mb, and dolomitic concretionary beds and/or nodules in the Vendomdalen Mb. In all three sections, bedding dips are normally 4°–8° in a SW direction; therefore, a fold test was not possible. In all, 37 horizons were collected from the DR section, 56 from the VR section, and 18 from the VM section, with some 15 m overlap between the VR and VM sections (Fig. 3). An approximate 3 m sampling gap occurs between the DR and VR sections, along with some larger sampling gaps (due to unsuitable lithologies) in the Vendomdalen Mb. The samples were collected by a water-cooled field drilling device and orientated using a dipmeter and magnetic compass.

Both thermal demagnetization and alternating demagnetization were utilized, using either a Magnetic Measurements thermal demagnetizer or a Molspin alternating field (AF) demagnetizer with a periodic reverse-tumbling action. Low-temperature demagnetization (LTD) was tested, but proved ineffective. With thermal demagnetization to temperatures higher than ~200 °C, heating-induced changes in the mineralogy often obscured the isolation of the characteristic remanent magnetization (ChRM). Consequently a composite demagnetization scheme using thermal demagnetization to 200 °C followed by AF demagnetization (to ~60–70 mT) was found to be generally most effective. At nearly all horizons two to four specimens were measured from each level. The specimens were measured on a CCL GM400 3-axis cryogenic magnetometer (noise level ~0.002 mA/m), using multiple sample positions, from which the magnetization variance was determined.

ChRM directions were isolated using the statistical procedures as implemented in LINEFIND, which is claimed to be superior to standard principal component analysis, since a balanced use is made of the measurement variance along with rigorous statistical procedures for identifying linear and planar structure in the demagnetization data (Kent et al., 1983). Both linear trajectory fits and great-circle (remagnetization-circle) data were used in defining the paleomagnetic behavior, guided by objective and qualitative selection of the excess standard deviation parameter (ρ), which governs how closely the model variance used for analysis matches the data measurement variance (Kent et al., 1983). The PMAGTOOL software was used for the analysis of mean directions.

MAGNETIC MINERALOGY

Progressive isothermal remanent magnetization (IRM) up to 3 T was applied to a representative subset of specimens, to investigate the magnetic mineralogy, which was also studied by low-temperature magnetic measurements (using a Princeton Measurements, Magnetic Property Measurement System [MPMS]) and reflected light microscopy.

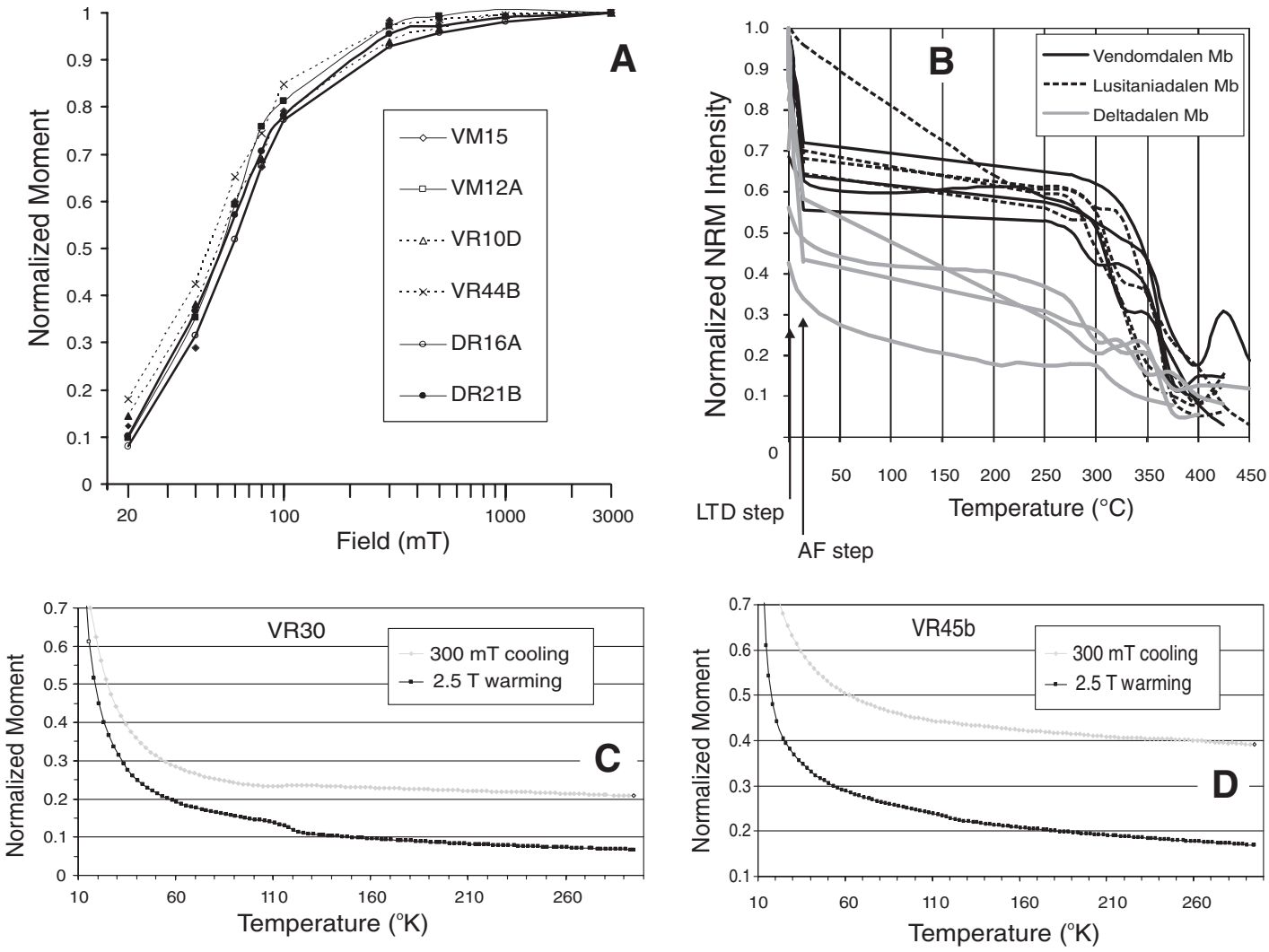


Figure 4. (A) Isothermal remanent magnetizations for six representative specimens. (B) Thermal demagnetization temperature versus the natural remanent magnetization (NRM) intensity, for representative specimens from the Vikinghøgda Formation (Fm). (C), (D) Low-temperature remanence measurements on two specimens (C-Lusitaniadalen Mb, D-Vendomdalen Mb), showing zero field cooling of a room temperature 300 mT isothermal remanent magnetization (IRM), and warming of a 2.5 T IRM applied at 10 K (normalized to moment at 10 K). Specimen stratigraphic location shown in Figure 8.

IRM acquisition experiments show a low-coercivity mineral, which largely saturates by 300 mT (Fig. 4A). This behavior could be due either to (titano) magnetite or a magnetic sulfide. LTD losses are between 5% and 23% showing that multidomain magnetite or titanomagnetite is present (Dunlop and Argyle, 1991; King and Williams, 2000). LTD losses are on average largest for the Deltadalen Mb and lowest for the Vendomdalen Mb. Demagnetization of a 2.5 T IRM applied at 10 K shows a weak magnetite Verwey transition for most samples and no evidence of the pyrrhotite magnetic transition at ~34 K (Figs. 4C and 4D). The small loss of moment on crossing the Verwey transition may be due to either the fine grain size (<~0.2 µm) or oxidation of the magnetite (King and Williams, 2000). Thermal demagnetization of the natural remanent magnetization (NRM) shows an accelerated loss of magnetization between ~300 °C and 370 °C in most specimens from the Vendomdalen and Lusitaniadalen members, a feature which is not as common in specimens from the Deltadalen Mb (Fig. 4B). The suspicion is that this behavior may be due to greigite(?) oxidation during heating (Roberts, 1995; Dekkers et al., 2000), although it could also relate to differences in blocking temperature characteristics of the magnetite remanence between the three members.

Optical (oil immersion), reflected-light microscopy up to a magnification of ×1000 was undertaken on nine representative samples to examine the sulfide mineralogy. Fe-Ti oxides identified (mostly >0.5 µm, the limit of optical resolution) were exclusively detrital magnetite or ilmenite. Textures of residual TiO_2 oxides indicate extensive dissolution of the originally deposited Fe-Ti oxides, which have been largely leached of Fe leaving relict TiO_2 oxides, typical of sediments subjected to reductive diagenesis (Morad and Aldahan, 1986; Hounslow et al., 1995). Residual Fe-Ti oxides occur in rock fragments, silicate grains, and some early-cemented patches. Sulfides are present in three forms: (1) early diagenetic framboidal pyrite; (2) equant pyrite, post-dating the framboidal pyrite; and (3) small amounts of an unidentified grayer colored (compared to pyrite), high-reflectivity sulfide phase (<10 µm in size). The abundance of this "gray phase" is unrelated to the amount

of late diagenetic equant pyrite present, and its association with the very early formed framboidal pyrite indicates that it probably formed early during burial. This "gray phase" does not appear to have the optical characteristics and ultra-fine grain size typical of greigite (R. Reynolds, 2006, personal commun.), and petrographically it is not pyrrhotite. It may be a later stage pyrite following framboidal pyrite grown but with a slightly different trace Ni/As composition. Small amounts of goethite from pyrite weathering are also present. In conclusion, the magnetization appears to be carried by a combination of magnetite (in part oxidized) and possibly greigite(?). The evidence for greigite is weak, in part due to the difficulty of clearly characterizing this phase without more extensive tests.

PALEOMAGNETIC RESULTS

Specimen demagnetization produced two magnetization components—firstly an often steep, downward-directed, overprint magnetization, which was sometimes removed at lower demagnetization stages (<200 °C) but often persisted into the early to mid stages (10–25 mT) of AF demagnetization (Fig. 5A). The directional dispersion of this component is notably larger for interpreted Triassic reverse- compared to normal-polarity specimens from the Deltadalen and Lusitaniadalen members and is also of different mean direction (Fig. DR2, Table DR1 [footnote 1]). This overprint component is interpreted to be composite, probably a Brunhes age overprint but combined with a variable amount of Triassic dual-polarity magnetization (Figs. 5A, 5C, and 5D). In 14% of specimens this composite-magnetization direction persisted until complete demagnetization. In the Vendomdalen Mb the overprint, when present, was often westerly or NW in direction (Figs. 5E and 5F; Fig. DR2).

Removal of this overprint revealed the ChRM component interpreted as a dual-polarity, NE- and SW-directed Triassic magnetization (Figs. 5A, 5B, and 6). This was often isolated above the 30–35 mT AF demagnetization stages, and was isolated through the origin in 49% of specimens, with the remainder being non-origin line fits due to hindrance from incomplete remanence cancellation at high-AF fields in the demagnetizer. Some 36% of specimens had suitable linear trajectory line fits (here termed "S-type" data). This S-type demagnetization behavior was visually classified into three quality classes—S1, S2, and S3 (Fig. 8B)—with mean α_{95} linear fits for these three data classes of 5.9°, 10.5°, and 16.5°, respectively. The mean excess standard deviation parameters for these classes are 1.3, 1.4, and 2.0, respectively, indicating the progressively larger model variance required to accommodate the line fits. The mean directions for the ChRM line fits in the DR section pass the reversal test (McFadden and McElhinney, 1990), whereas those for the VR section fail (Table 1; Fig. 6A).

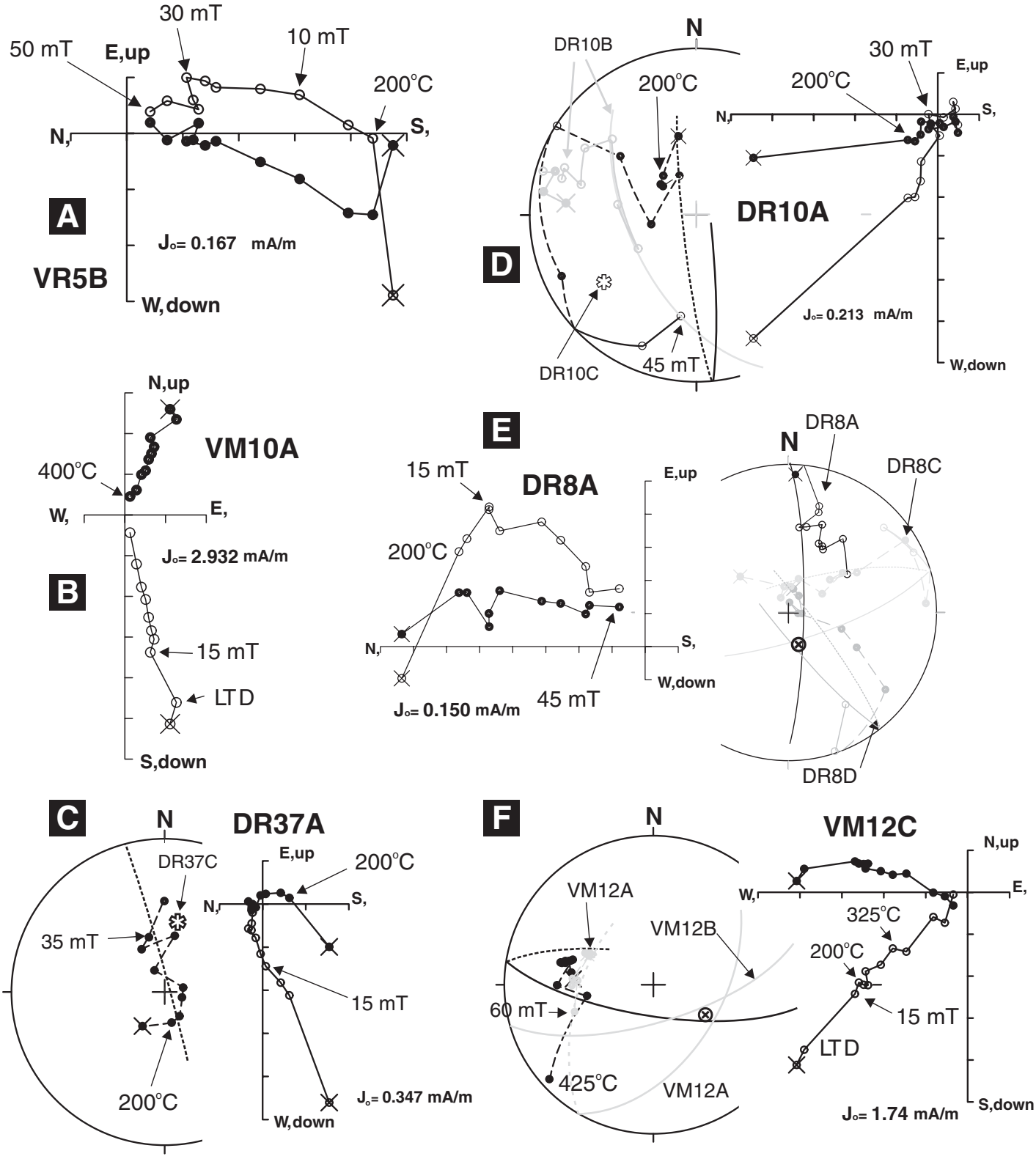
Some 49% of specimens displayed great-circle trends, of varying arc length, toward interpreted Triassic reverse- and normal-polarity directions (here referred to as T-type demagnetization behavior; Figs. 5C–5F). This T-type behavior was visually classified into three quality classes—T1, T2, and T3 (Fig. 8B)—based on the visual length and scatter of the demagnetization points about the great circle, with T1 being the best quality. The mean α_{95} for the poles to the fitted planes, for these three data classes are 17.3°, 18.5°, and 20.0°, respectively (mean ρ of 1.4, 1.6, 1.7, respectively). These great-circle fits included the origin in 91%, 83%, and 81% of the specimens with T1, T2, and T3 classes, respectively.

Mean Directions and Paleopoles

It is possible to determine mean directions for the majority of the horizons with multiple successful specimens per horizon, using the methods of McFadden and McElhinney (1988). However, using such "horizon means" (see GSA Data Repository Table DR2) to determine member mean directions utilizes horizon means often defined by only two to three specimens, which is an inadequate number to properly represent the true secular variation. Consequently, member and section means were determined using "specimen-based" means, combining the great-circle paths with the specimen line-fit ChRM (Table 1), to determine combined means using the method of McFadden and McElhinney (1988). This method determines a mean direction, by including the "fixed-point" ChRM directions, and those points on the projected great circles, which maximize the resultant length (i.e., in effect, those calculated points on the great circle that are closest to the combined mean direction). This method tends to slightly underestimate the confidence interval (McFadden and McElhinney, 1988). This method of specimen-based, mean-direction determination is similar to most other magnetostratigraphic studies, which have a limited number of specimens (often only one) from each sampled horizon.

The mean axial direction through the poles to the great-circle paths (i.e., pole to the mean girdle plane) can also provide an estimate of the unresolved magnetization component (Van der Voo, 1993), without the ChRM fixed-point constraints. In these data at member level, this

Figure 5. Representative specimen demagnetization data, specimen location on Figure 8. (A) VR5B, reverse polarity. Origin line-fit characteristic remanent magnetization (ChRM) isolated from 30 mT (180°, –33°, α_{95} = 10°, demagnetization class S3), following single thermal demagnetization step at 200 °C. (B) VM10A—normal polarity. ChRM isolated between 250 °C and 400 °C (028°, +64°, α_{95} = 6°, class S1) following low-temperature demagnetization (LTD) and a single alternating field (AF) demagnetization step. NRM— natural remanent magnetization. (C) DR37A—normal polarity. Strong composite overprint component (148°, 61°, α_{95} = 6°) isolated between 5 and 35 mT demagnetization (following thermal demagnetization at 200 °C, class T2). No ChRM isolated, but origin intersecting great-circle plane (255°, +3°, α_{95} = 16°) fitted from NRM value. Other specimen at this horizon (DR37C), produces an origin-intersecting ChRM (010°, 52°, α_{95} = 11°), shown with asterisk (*). (D) DR10A, interpreted reverse-polarity–fitted, great-circle plane from NRM to 45 mT (264°, –7°, α_{95} = 20°, class T1). Other specimens at this horizon produce an origin intersecting line-fit ChRM from 40 mT (specimen DR10C, 234°, –33°, α_{95} = 18°, class S3), and origin-intersecting, great-circle fit from 35 mT (specimen DR10B, 246°, 34°, α_{95} = 26°). (E) DR8A, interpreted reverse polarity. Fitted great-circle plane from NRM to 35 mT (271°, –9°, α_{95} = 11°; class T1). Demagnetization points and fitted great-circle planes also shown for sister specimens 8C (NRM-origin; 344°, –16°, α_{95} = 11°, class T2) and 8D (NRM-origin, 051°, –9°, α_{95} = 11°, class T2). The great-circle analysis produces a mean direction for this level of 161, –72 (marked with x). (F) VM12c, interpreted reverse polarity. Strong composite overprint component (291°, +34°, α_{95} = 4°) isolated between 15 mT and 400 °C demagnetization (following low-temperature demagnetization [LTD]). Origin intersecting great-circle plane fitted from 15 mT to 425 °C (009°, –17°, α_{95} = 23°, class T3). Points and fitted great-circle data for sister specimen VM12A (NRM –50 mT, 164°, 21°, α_{95} = 26°, class T3), and fitted great circle for specimen VM12B (15 mT –425 °C, 009°, –17°, α_{95} = 23°, class T3) shown in gray. The great-circle analysis produces a mean direction for this level of 115°, –52° (marked with x). Lower projection paths dashed; upper projection paths are solid.


A
VR5B
50 mT
30 mT
10 mT
E,up
200°C
N,
S,
W,down
Jo= 0.167 mA/m
B
VM10A
N,up
400°C
W,
E,
15 mT
LTD
S,down
Jo= 2.932 mA/m
C
DR37A
N
DR37C
E,up
200°C
N,
S,
35 mT
15 mT
200°C
W,down
Jo= 0.347 mA/m
D
DR10B
N
200°C
DR10C
45 mT
DR10A
30 mT
E,up
200°C
N,
S,
W,down
Jo= 0.213 mA/m
E
DR8A
15 mT
E,up
200°C
N,
S,
45 mT
W,down
Jo= 0.150 mA/m
N
DR8A
DR8C
DR8D
F
VM12C
N
VM12A
VM12B
60 mT
425°C
VM12A
N,up
W,
E,
325°C
200°C
15 mT
LTD
S,down
Jo= 1.74 mA/m

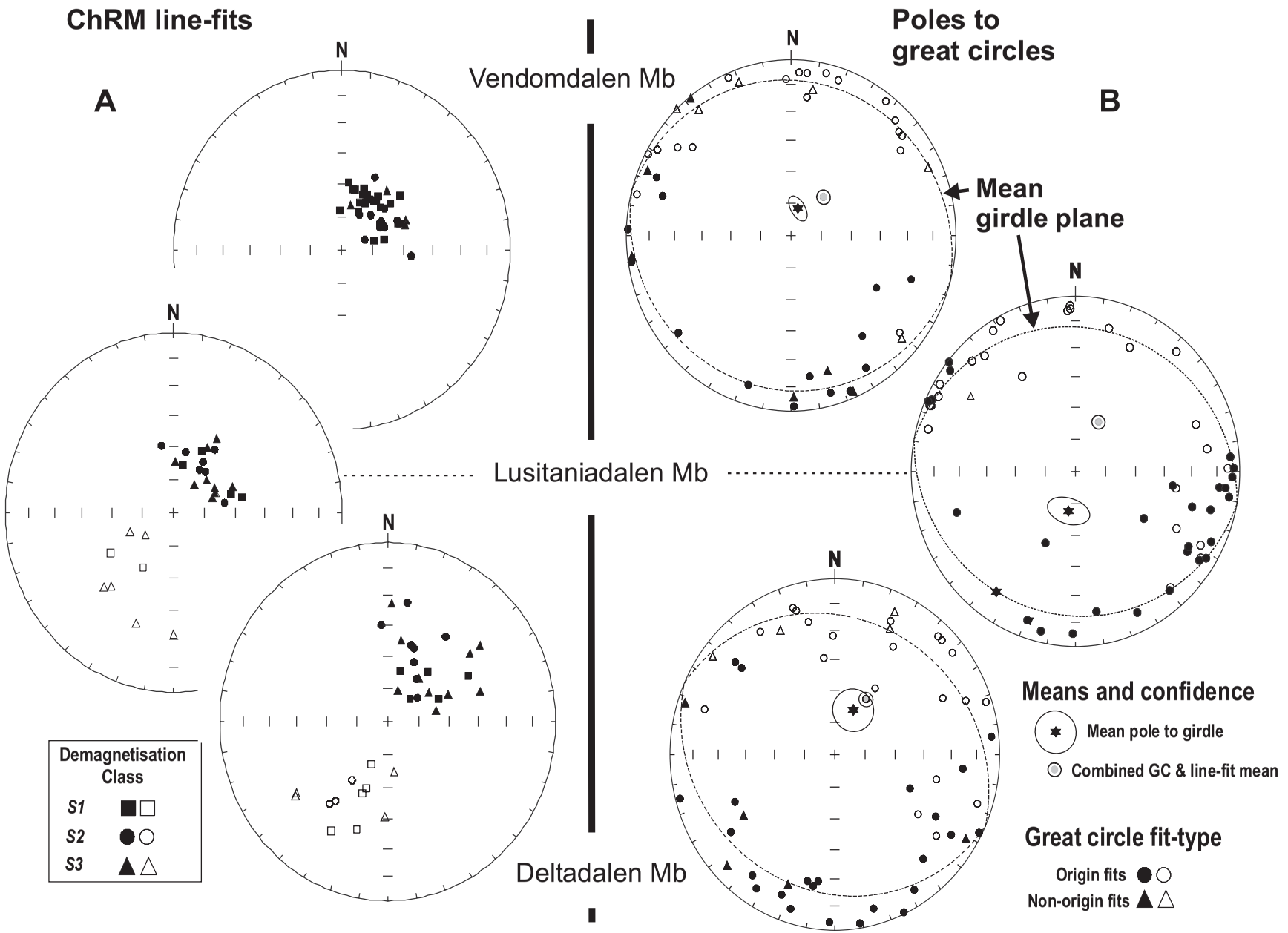


Figure 6. (A) Stereographic projection of the characteristic remanent magnetization (ChRM) directions (i.e., S-class data) from the Vendomdalen, Lusitaniadalen, and Deltadalen members. The quality of the ChRM line fits are classed into three groups, with S1 the best quality, and S3 the least (see text for details). (B) Poles to the specimen-fitted, great-circle planes for the three members, along with the combined ChRM–great-circle mean, and its Fisher confidence cone. The girdle plane (dotted) is the mean plane through the great-circle poles, along with the axis (filled star) through the mean plane, and its elliptical 95% confidence region (using statistics in Fisher et al., 1987). Filled symbols—lower hemisphere; open symbols—upper hemisphere.

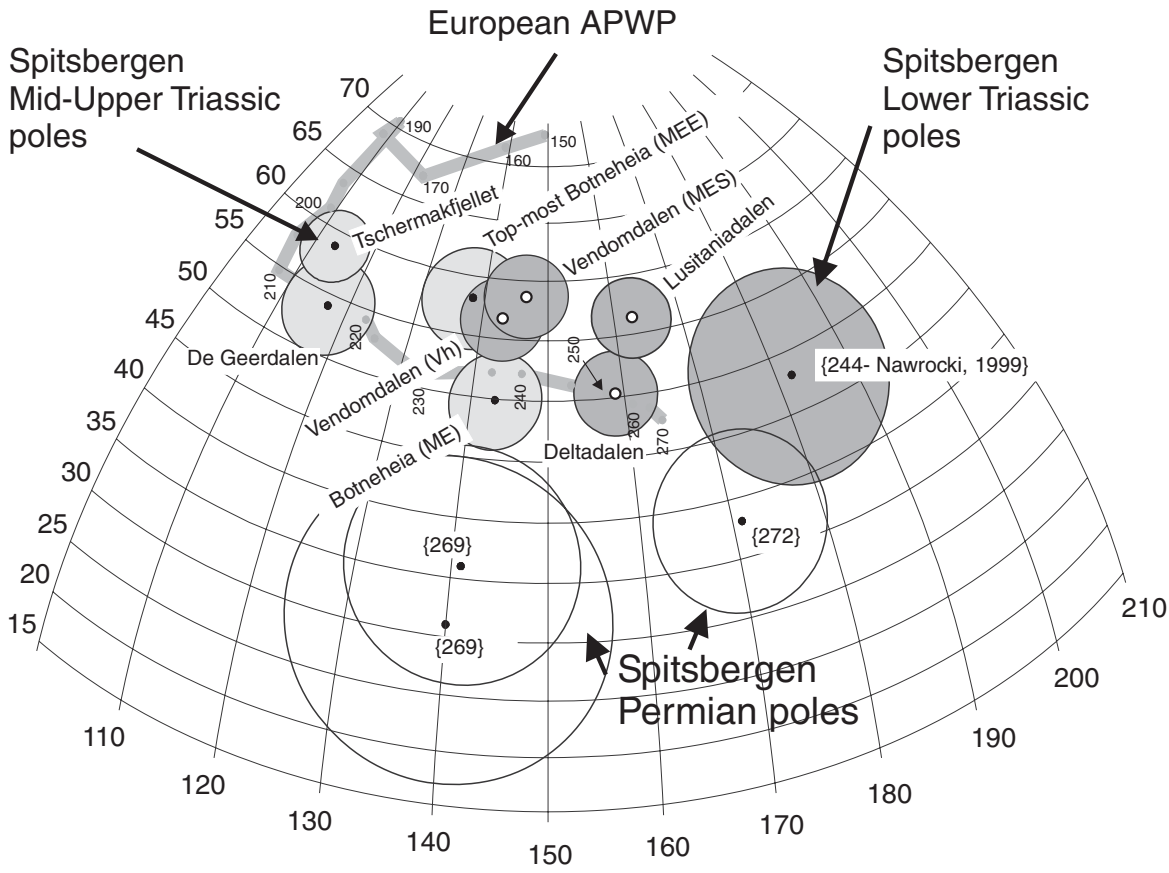


Figure 7. The mean virtual geomagnetic pole (VGP) directions for the Deltadalen, Lusitaniadalen, and Vendomdalen members (with 95% confidence cones), compared with other relevant VGP data from Spitsbergen. Light-gray filled confidence cones are data from the Spitsbergen Middle and Upper Triassic (Botneheia Fm, Tschermakfjellet and De Geerdalen formations) from Hounslow et al. (2007a, 2008). Dark-gray filled confidence cones are data from the Spitsbergen Lower Triassic (this study, Hounslow et al., 2008; Nawrocki, 1999). MEE, ME, and MES data from Milne Edwardsfjellet (Hounslow et al., 2008; Fig. 2). Thick gray line—mean European apparent polar wander path (Torsvik et al., 2001), with age of poles indicated in 10 Ma increments from 270 Ma. Unfilled 95% confidence cones are data from the Permian of Svalbard, labeled with age {…} in million years (from Torsvik et al., 2001). APWP—Apparent polar wander path.

TABLE 1. DIRECTIONAL MEANS, REVERSAL TESTS, AND VIRTUAL GEOMAGNETIC POLES

Type and/or unit	Declination	Inclination	K	α_{95}	Nl/Np	Reversal test	G_O/G_C	P_{lat}	P_{long}	Dp/Dm
DR section										
Line fits$	35.3	54.6	17.3	6.3	32/0	Rb	9.6/13.9*	44	156	6.3/8.9
Great circle means+	35.2	59.1	21.5	3.5	32/47	Ra	3.0/8.8*	49	154	4.0/5.4
VR section										
Line fits$	38.4	62.2	26	4.2	47/0	R-	16/13.6	52	150	5.1/6.5
Great circle means+	27.3	63.5	23.1	3.0	47/61	Rb	9.9/11	55	162	3.7/4.7
VM section										
Line fits$	39.4	64.6	61.6	3.6	26/0	-	-	55	147	4.6/5.8
Great circle means+	39.3	66.3	43.7	3.9	26/6	R-	38/23*	57	147	5.3/6.4
Vendomdalen Fm+	42.4	66.2	42.4	3.3	40/6	R-	31/11*	57	143	4.4/5.4
Lusitaniadalen Fm+	26.4	64.3	24.5	3.2	28/53	Rb	5.5/12	56	163	4.4/5.4
Deltadalen Fm+	30.7	59.8	20.1	3.4	35/53	Rb	5/8.6*	50	159	3.9/5.1

Note: +—Great-circle combined mean using method of McFadden and McElhinny (1988); $—conventional Fisher mean; Nl—number of specimens using with fitted lines; Np—number of specimens with great-circle planes used in the determination of the mean direction; GC—Great Circle; α_{95}—Fisher 95% cone of confidence; and K—Fisher precision parameter. G_O is the angular separation between the inverted reverse and normal directions, and G_C is the critical value for the reversal test. In the reversal test, the G_O/G_C values flagged with * indicate common K values; others not flagged have statistically different K values for reverse and normal populations, in which case a simulation reversal test was performed. Ra, Rb indicates a pass of the reversal test with G_O <5 and <10, respectively. R- is a failed reversal test. P_{lat} and P_{long} are the latitude and longitude of the mean virtual geomagnetic pole (VGP). Section location: 78.26°N, 16.87°E.

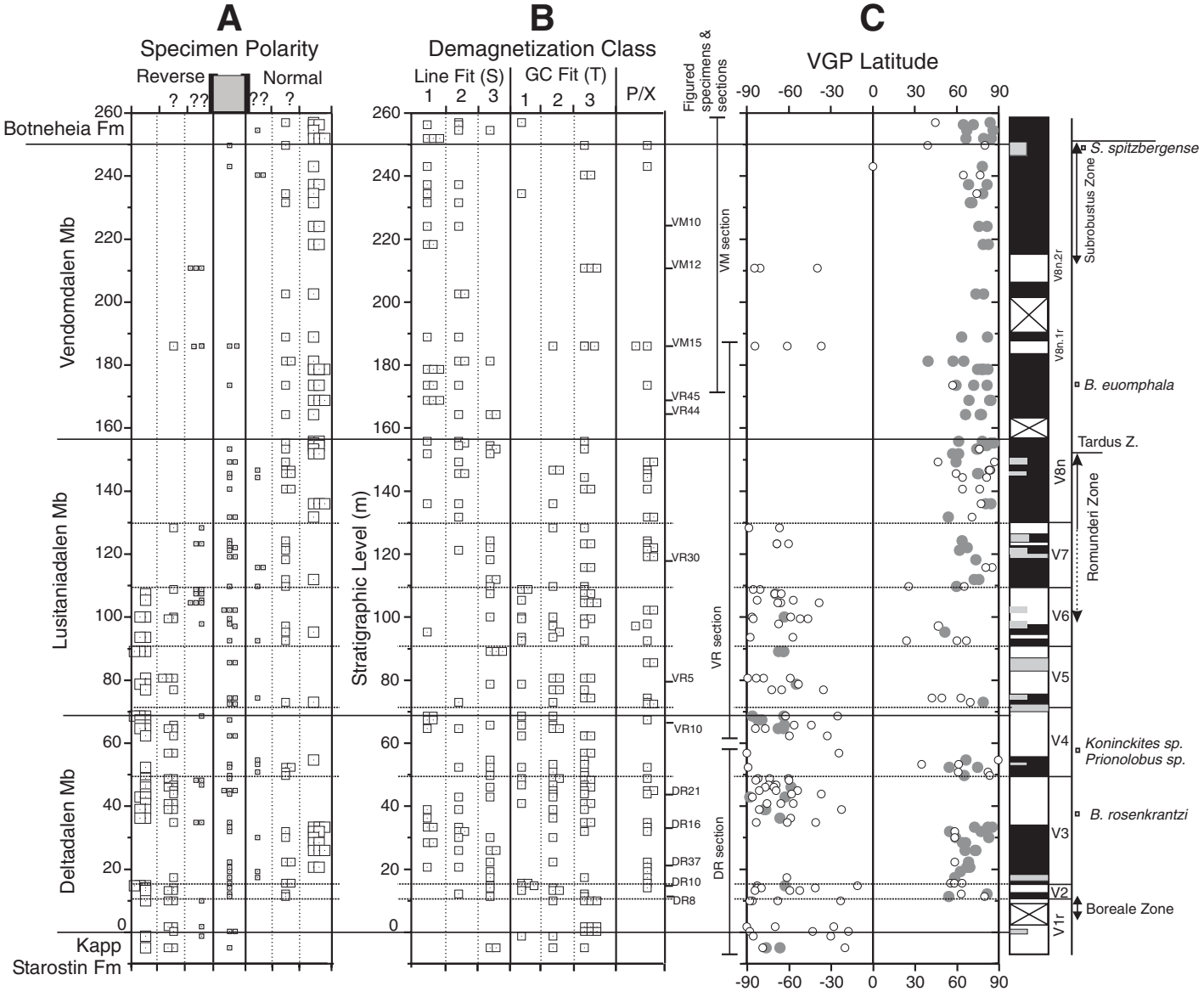


Figure 8. Magnetostratigraphic and biostratigraphic data summary. (A) Interpreted specimen polarity quality, with those in the mid-gray column not assigned a polarity. Poorest quality in column headed with question marks (??); best in column with largest symbols. (B) Demagnetization behavior showing categorization into good (S1) and poor (S3) characteristic remanent magnetization (ChRM) line fits; great-circle fit quality range from good (T1) to poor (T3), and specimens with no Triassic magnetization are indicated in the P/X column (see text for details). The right side shows the location of mineral magnetic specimens and those in Figure 5. (C) The virtual geomagnetic pole (VGP) latitude, with filled symbols for those specimens possessing an S-class ChRM, and unfilled symbols for specimens with T-class great-circle behavior. White—reversed polarity, black—normal polarity, gray—uncertain. Half bar-width indicates a single useful specimen from this horizon. Dotted lines are magnetozone boundaries.

mean axial direction is seen to be displaced to a steeper inclination, probably reflecting the in part composite nature of the unresolved magnetization (Fig. 6B). Hence, either the ChRM directions alone, or the combined great-circle–ChRM directions represent the best estimate of the mean field direction (Fig. 6B). The combined mean directions for the Deltadalen and Lusitaniadalen members pass the reversal test (McFadden and McElhinny, 1990) with classes Ra and Rb, respectively (Table 1). The specimen data from the Vendomdalen Mb fail the reversal test because of the lack of any line-fit ChRM data to constrain the reverse combined mean and the low number of reverse-polarity specimens.

The paleopoles for the sections and the members fall near to the Lower Triassic segment of the European apparent polar wander path with the mean reported by Hounslow et al. (2008) for the Vendomdalen Mb being similar to this member at Vikinghøgda (Fig. 7). Overall, the available Triassic paleopole data (and Permian data; Nawrocki, 1999) from Spitsbergen show a close similarity to the westward track of the European apparent polar wander path (APWP) (Fig. 7). This is entirely compatible with hydrocarbon exploration results from the Barents Sea (Stemmerik and Worsley, 2005), which indicate that Spitsbergen is the most NW part of the Europe-Baltica craton. Little significance can be attached to the Lower Triassic paleopole derived from the Trygghamna section in western Spitsbergen (Nawrocki, 1999), which is based on only eight samples (Fig. 7). Taken at face value, it suggests some ~25° (±14°) counterclockwise tectonic rotation of the Trygghamna section.

MAGNETOSTRATIGRAPHIC INTERPRETATION

The line-fit ChRM directions from Vikinghøgda were converted to virtual geomagnetic pole (VGP) latitude using the great-circle combined mean directions from each respective section (Table 1 and Fig. 8C). For those specimens that had no line fit, the point on their great circle nearest the combined mean was used for calculating the VGP latitude (Fig. 8C). This calculated point (constrained by each "remagnetization" great circle) is actually the direction used in determining the combined mean direction (McFadden and McElhinney, 1988). All specimens were also assigned a "polarity quality" (Fig. 8A) based on the quality of demagnetization behavior and, if from T-class specimens, the length and end-point position of the great-circle trend (similar to the procedures used by Ogg and Steiner, 1991; Hounslow et al., 2007a). One specimen of good-quality polarity (i.e., S-type) was sufficient to define the horizon polarity, whereas with specimens of poorer quality, at least two were needed (Figs. 3 and 8A). Some 20% of specimens failed to yield data that could be used to determine horizon polarity, and five horizons failed to yield any specimens that could reliably be used to determine magnetic polarity (Figs. 3 and 8). The magnetostratigraphy can also be evaluated using the great-circle and Fisherian directional means at the horizon level, which produces a horizon mean-based magnetostratigraphy very similar to that based on the individual specimen classifications (Table DR2).

Major magnetozone reverse and normal couplets have been numbered (Figs. 3 and 8C) from the base of the section, using the prefix Vh (for Vikinghøgda). The magnetic polarities of six magnetozones are defined with multiple specimens from a single sampling horizon (Vh6n.1n, Vh6n.1r, Vh7n.1r, Vh7r, Vh8n.1r, and Vh8n.2r). The overlap between sections VR and VM shows consistent polarity (Figs. 3 and 8C).

Magnetic Polarity Records from Spitsbergen

The magnetostratigraphy of the Vendomdalen Mb and overlying Botneheia Fm at Milne Edwardsfjellet (Figs. 2 and 9) has been detailed by Hounslow et al. (2007a, 2008). The magnetostratigraphy demonstrates two substantive reverse magnetozones—MF1r in the mid parts of the Vendomdalen Mb and MF2r–MF3r in the transition from the Vendomdalen Mb into the Botneheia Fm (Fig. 9). Magnetozone interval MF2r–MF3r is not represented at Vikinghøgda, due to an interpreted erosional disconformity at the lower boundary of the Botneheia Fm. Additional evidence for this is: (a) absence of a well-represented coarsening-up succession at the top of the Vendomdalen Mb at Vikinghøgda, a feature which is regionally typical, and is seen at Milne Edwardsfjellet; (b) the *Rhizocorallium*-bearing bed at 257 m in the section (Fig. 3) is lithologically equivalent to a bed some 30 m above the base of the Botheneia Fm at Milne Edwardsfjellet (Hounslow et al., 2008); (c) reworked late Permian miospores occur in the base of the Botneheia Fm at Milne Edwardsfjellet. Evidently erosion did not remove the entire uppermost part of the Subrobustus Zone at Vikinghøgda, since its uppermost subzone is represented by the ammonoid *S. spitzbergense* (Figs. 8 and 9). The boundary beds between the Vendomdalen and Botneheia Fm are characterized by a thin normal magnetozone (MF3n at Milne Edwardsfjellet), at the very top of which occurs the ammonoid *Karangatites evolutus* (Hounslow et al., 2008) indicating the second subzone of the Boreal Anisian. The base Anisian is here informally used as the first occurrence of the conodont *Chiosella timorensis* in the Dęsli Caira section (Grădinaru et al., 2007), a level which can be confidently correlated to the mid parts of MF4n.1r (Hounslow et al., 2007b; Fig. 10, see later). The conodont *C. timorensis* occurs in the Milne Edwardsfjellet section in the MF4r magnetozone (Fig. 9).

The ammonoids from Milne Edwardsfjellet suggest the lowest three zones of the Spathian are there condensed into some 20 m, whereas the presence of *B. euomphala* at 17 m above the base of the Vendomdalen Mb at Vikinghøgda indicates a thicker interval of lower Spathian (Fig. 9). This relationship suggests that magnetozone Vh8n.1r is the equivalent of MF1n.2r (Fig. 9). We argue later that Vh8n.2r is probably equivalent to MF2n.2r, rather than MF1r.

Nawrocki and Grabowski (2000) determined a magnetostratigraphy from Tschermakfjellet and Kapp Wijk in Dicksonland and Trygghamna in western Spitsbergen, with a rather wide sample spacing (Figs. 2 and 9). Using logs of equivalent and nearby sections to those studied by these authors, we are able to approximately map the lithostratigraphy of Mørk et al. (1999a, 1999b) onto the sections studied by Nawrocki and Grabowski (2000). These lithostratigraphic constraints support possible magnetostratigraphic correlations to the Vikinghøgda and Milne Edwardsfjellet sections (Fig. 9). The two reverse-polarity magnetozones (i.e., TF4r and TF5r) in the upper part of the Vendomdalen Mb at Tschermakfjellet (Fig. 9) probably are equivalents of magnetozones MF1r and MF2r at Milne Edwardsfjellet; although it is also possible that TF4r may be the equivalent of MF2n.2r. The presence of the Smithian is indicated by ?*Arctoceras* sp. some 68 m below the base of the Botneheia Fm at Tschermakfjellet, along with Romunderi Zone and Tardus Zone faunas from the Dicksonland region (Tozer and Parker, 1968). Korchinskaya (1986) has also found at different stratigraphic levels, *B. rosenkrantzi* and *V. spitsbergensis,* in the Dicksonland sections, indicating the late Griesbachian and Dienerian.

The *Myalina*-bearing sandstone in the upper part of the section at Trygghamna (Fig. 9) can be correlated to a similar interval in the Bellsund and Hornsund areas of western Spitsbergen, which contains conodonts (including *Ns. svalbardensis*) of mid-Dienerian to early Smithian age (Nakrem and Mørk 1991). In addition, *Otoceras boreale* occurs in the Selmaneset Mb at Festningen (Wignall et al., 1998) along with a more diverse Griesbachian fauna from the Bellsund area in western Spitsbergen (Nakazawa et al., 1990). These constraints suggest the normal magnetozone in the upper part of the Selmanest Mb at Trygghamna is the probable equivalent of Vh4n (Fig. 9), which would place the occur-

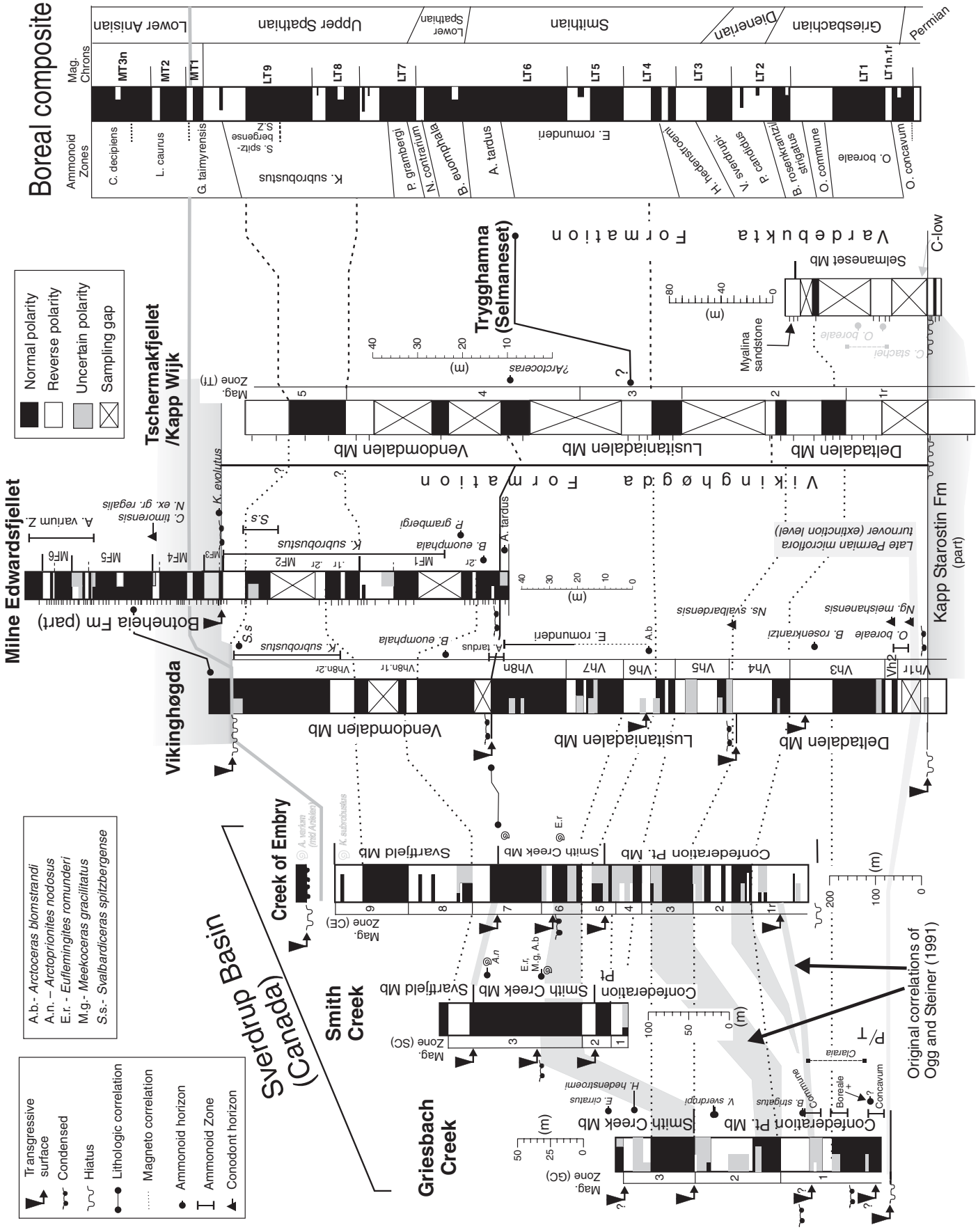

Boreal composite
Mag. Chrons
Ammonoid Zones
Lower Anisian
Upper Spathian
Lower Spathian
Smithian
Dienerian
Griesbachian
Permian
MT3n
MT2
MT1
LT9
LT8
LT7
LT6
LT5
LT4
LT3
LT2
LT1
LT1n.1r
C. decipiens
L. caurus
G. taimyrensis
K. subrobustus
S. spitz-bergense S.Z
P. grambergi
N. contrarium
B. euomphala
A. tardus
E. romunderi
H. hedenstroemi
V. sverdrupi-candidus
P. candidus
B. fosse-strigatus
O. commune
O. boreale
O. concavum
Normal polarity
Reverse polarity
Uncertain polarity
Sampling gap
Milne Edwardsfjellet
Botneheia Fm (part)
Tschermakfjellet /Kapp Wijk
Vikinghøgda
Trygghamna (Selmaneset)
?Arctoceras
Myalina sandstone
Selmaneset Mb
C-low
Vardebukta Formation
Vendomdalen Mb
Lusitaniadalen Mb
Deltadalen Mb
Vikingbøgda Formation
Kapp Starostin Fm (part)
Late Permian microflora turnover (extinction level)
Mag. Zone (Tr)
MF6
MF5
MF4
MF3
MF2
MF1
Vhn.2r
Vhn.1r
Vhn
Vh7
Vh6
Vh5
Vh4
Vh3
Vh2
Vh1r
A. varium Z.
C. timorensis
N. ex. gr. regalis
K. evolutus
S.s.
K. subrobustus
P. grambergi
B. euomphala
A. tardus
E. romunderi
A.b
O. boreale
B. rosenkrantzi
Ns. svalbardensis
Ng. meishanensis
Creek of Embry
Smith Creek
Griesbach Creek
Sverdrup Basin (Canada)
Svartfjeld Mb
Smith Creek Mb
Confederation Pt. Mb
Mag. Zone (CE)
Mag. Zone (SC)
Mag. Zone (GC)
A.n
E.r, M.g, A.b
E.r
H. hedenstroemi
E. circratus
V. sverdrupi
B. strigatus
O. commune
Boreale
Concavum
Clarkia
P/T
Original correlations of Ogg and Steiner (1991)
A.b.- Arctoceras blomstrandi
A.n. – Arctoprionites nodosus
E.r. - Euflemingites romunderi
M.g.- Meekoceras gracilitatus
S.s.- Svalbardiceras spitzbergense
Transgressive surface
Condensed
Hiatus
Lithologic correlation
Magneto correlation
Ammonoid horizon
Ammonoid Zone
Conodont horizon

Figure 9. Magnetostratigraphy of other Boreal Lower Triassic sections and their correlation to the Spitsbergen data. The Sverdrup Basin polarity columns have been slightly modified, using the Ogg and Steiner (1991) published data, to give a more optimistic polarity interpretation. Biostratigraphy datums in the Canadian sections from (Ogg and Steiner, 1991; Tozer, 1994; Henderson and Baud, 1997). Milne Edwardsfjellet section from Hounslow et al. (2008), Tschermakfjellet/Kapp Wijk and Trygghamna sections from Nawrocki and Grabowski (2000). Ammonoid data from Buchan et al. (1965), Nakazawa et al. (1990), and Wignall et al. (1998). Magnetozones in the Sverdrup Basin data have been relabeled according to the section (GC, SC, and CE), rather than according to interpreted age, as originally used by Ogg and Steiner (1991). Magnetozones in the Boreal composite have been labeled LT for the Lower Triassic and MT starting in the Middle Triassic. Base Anisian used is the first occurrence of *Chiosella timorensis* in the Dęsli Caira section (Grădinaru et al., 2007; see Fig. 10).

rence of *O. boreale* at Festningen into the Commune Zone, a feature also described from East Greenland (Bjerager et al., 2006).

SYNTHESIS OF A BOREAL LOWER TRIASSIC BIOMAGNETOSTRATIGRAPHY

The magnetostratigraphic and biostratigraphic data from the three Sverdrup Basin sections studied by Ogg and Steiner (1991) can be integrated with the Spitsbergen data to produce a Boreal composite biomagnetostratigraphy for the entire Lower Triassic (Fig. 9). The sections studied by Ogg and Steiner (1991) at Smith Creek and Creek of Embry are within 2 km of each other (on Ellesmere Island), with intersection correlation provided by four marker beds and several ammonite-bearing horizons (Fig. 9). These correlations were supported by the magnetostratigraphy, except that the uppermost reverse magnetozone in the Smith Creek Mb (CE6r; Fig. 9), at the Creek of Embry, was not detected at the Smith Creek section.

The biostratigraphy for the Griesbach Creek section (NW Axel Heiberg Island, 130 km from Smith Creek) originally presented by Ogg and Steiner has been modified slightly using the data from Tozer (1994) and Henderson and Baud (1997), particularly the occurrence of the basal Smithian zonal index ammonoid *Hedenstroemia hedenstroemia*, some 25 m below *Euflemingites cirratus* (indicating the Romunderi Zone; Tozer, 1994) in the upper part of the Smith Creek Mb.

In addition to the magnetostratigraphy and biostratigraphy, the sequence stratigraphy from the Sverdrup and Svalbard Boreal successions provides a supporting framework for correlation (Mørk et al., 1989). In their study of the Sverdrup Basin sections Ogg and Steiner (1991) provided both a sequence stratigraphic interpretation, along with horizons flagged as condensed and horizons of hiatus. Their sequence stratigraphy largely follows the basin-wide synthesis of Embry (1986), in which basin-margin sandstone tongues (the Cape Butler, Pell Point, and Cape O'Brien members) are reflected in shale-dominated, but coarsening-up units in the basin center. They also identify additional transgressive surfaces (and sequences) within the Confederation Point and Smith Creek members in their basin-center sections (Fig. 9). Within the Confederation Point and Smith Creek members, Ogg and Steiner (1991) associate intervals of hiatus with the upper-regressive parts of some of these coarsening-up units, which they suggest are at ammonite-rich "condensation horizons." This interpretation is questionable, in that: (a) biostratigraphy for the most part is not sufficiently continuous to confirm hiatus; (b) the boundaries of the members in the Sverdrup Basin Lower Triassic are conformable even at basin margins (Embry, 1986); and (c) ammonoid preservation is strongly controlled by early cementation, which is not necessarily related to condensation or hiatus. Intervals of condensed sedimentation were also identified by Ogg and Steiner (1991) following some transgressive surfaces, and in the mid parts of the Smith Creek Mb (Fig. 9). Those intervals of condensed sedimentation between the transgressive surface and the maximum flooding surface probably have utility in understanding possible changes in sedimentation rate related to the sequence stratigraphy, and are retained on Figure 9. In addition to the transgressive surfaces indicated by Ogg and Steiner (1991), regional transgressive surfaces also occur at the base of the Confederation Point and Svartfjeld members (Embry, 1986; Mørk et al., 1989).

Clear evidence of hiatus is present at two of these Sverdrup Basin sections. Firstly, at Greisbach Creek, the Confederation Point Mb (with *O. concavum)* rests on Artinskian (Lower Permian) strata (Henderson and Baud, 1997); secondly, the top of the Svartfjeld Member in the Creek of Embry section, containing *K. subrobustus,* is abruptly overlain by strata with mid-Anisian ammonoids (Ogg and Steiner, 1991; Fig. 9).

Induan

Magnetozone Vh3n is interpreted as the upper part of the normal magnetozone (composite magnetozone LT1n; Fig. 9) widely reported from the lower part of the Griesbachian (Heller et al., 1988, 1995; Steiner et al., 1989; Gallet et al., 2000; Scholger et al., 2000; Steiner, 2006). However, the lowest part of the Deltadalen Fm, coincident with *O. boreale*, is characterized by reversed polarity in magnetozones Vh1r and Vh2r, with a 6 m sampling gap in Vh1r. Using supporting evidence from the thicker Permian-Triassic boundary (PTB) sections in Tethyan marine sections (discussed below), we suggest that a short reverse magnetozone (LT1n.1r) occurs within LT1n, coincident with magnetozone Vh2r at Vikinghøgda (Figs. 9 and 10). In spite of a comparable sample spacing to ours, Ogg and Steiner (1991) failed to detect a reverse-polarity interval within LT1n at Griesbach Creek. However, four of their 21 sample levels in LT1n produced uncertain polarity, with one level that had a negative inclination and a second level that had a declination with a 180° swing from the expected normal-polarity declination. It is perhaps possible that one or the other of these levels may represent LT1n.1r but are not properly resolved due to the overprint in the Griesbach Creek section. The difficulty of resolving the reverse-polarity directions in the Griesbach Creek section is clearly evident by the large number of uncertain polarity intervals within parts of the section that are probably reverse polarity, based on correlation to the other Boreal sections (Fig. 9).

Ogg and Steiner (1991) utilized two constraints to provide correlation between the Ellesmere Island sections and the Griesbach Creek section. The first constraint is the regional conformity of the boundary between the Smith Creek and Confederation Point members. However, this lithostratigraphic correlation is violated by *H. hedenstroemia* in the uppermost reverse magnetozone (magnetozone GC3r, Fig. 9) at Griesbach Creek, which, using their constraint, would occur above the Romunderi Zone fauna at Smith Creek. Hence, the lower part of the Smith Creek Mb at Griesbach Creek must be older than that at Smith Creek and Creek of Embry. Secondly, Ogg and Steiner correlated the transgressive surface boundary at 65.3 m in the Griesbach Creek section, with a similar surface at ~80 m at Creek of Embry (Fig. 9). Using the revised constraints provided by *H. hedenstroemia* suggests the substantive normal magnetozone in the lower part of the Smith Creek Mb at

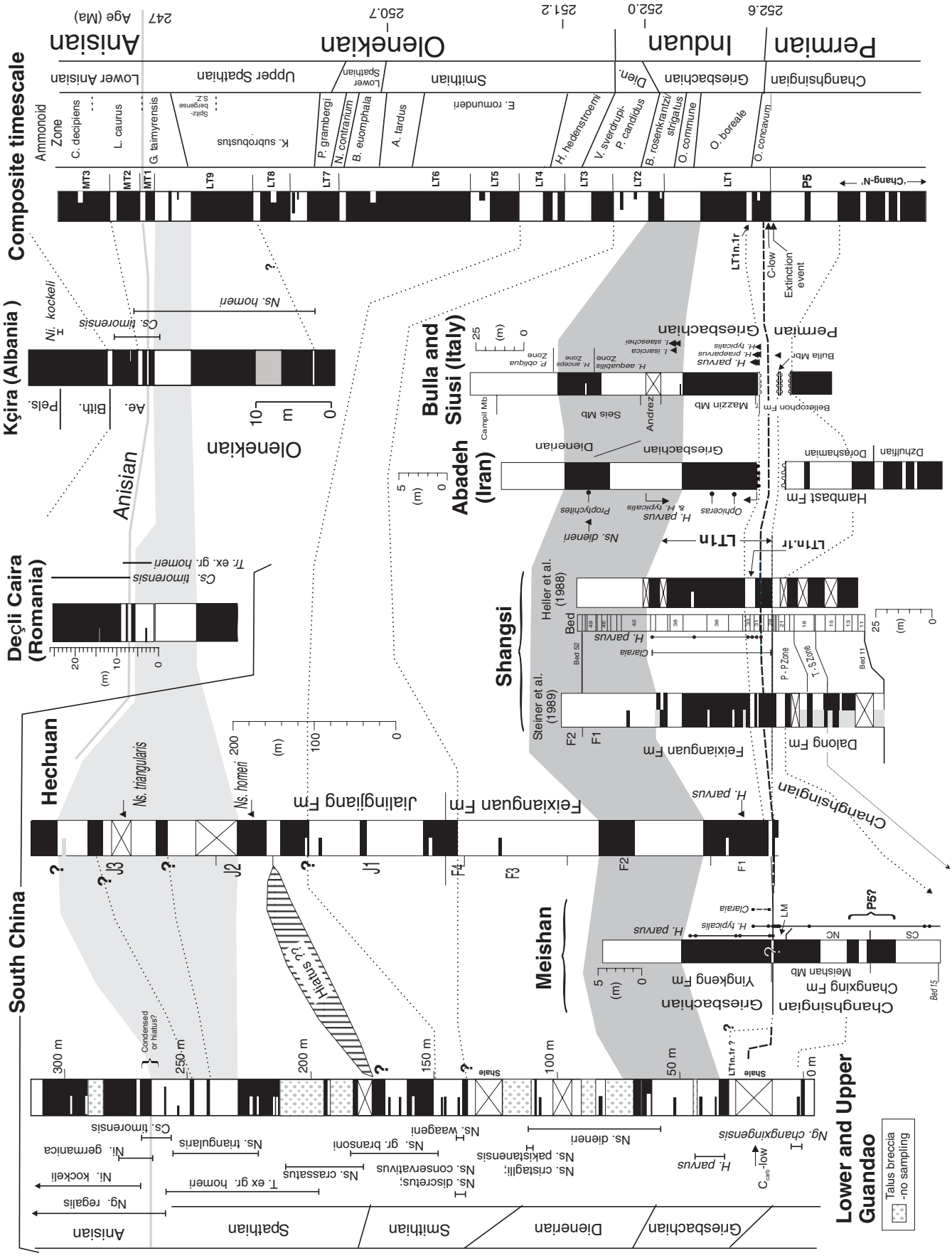

Composite timescale
Permian
Induan
Olenekian
Anisian
Changhsingian
Griesbachian
Dien.
Smithian
Spathian
Lower Anisian
Age (Ma)
252.6
252.0
251.2
250.7
247
Ammonoid Zone
C. deciipiens
L. caurus
G. taimyrensis
K. subrobustus
E. romundeni
H. hedenstroemi
V. sverdrupi-P. candidus
B. rosenkrantzi/ strigatus
O. commune
O. boreale
O. concavum
P. grambergi
N. contratum
B. euomphala
A. tardus
LT1n.1r
C-low
Extinction event
Kçira (Albania)
Ni. kockeli
Cs. timorensis
Ns. homeri
Ae.
Bith.
Pels.
Olenekian
Anisian
Deçli Caira (Romania)
Cs. timorensis
Tr. ex gr. homeri
Bulla and Siusi (Italy)
Permian
Griesbachian
Bulla Mbr
Mazzin Mb
Seis Mb
Campil Mb
Andrez
Bellerophon Fm
H. parvus
H. praeparvus
H. typicalis
H. aequabilis Zone
H. anceps Zone
P. oblqua Zone
Abadeh (Iran)
Griesbachian
Dienerian
Griesbachian
Hambast Fm
Dzhulfian
Dorashamian
Ophiceras
H. parvus & H. typicalis
Propyrchites
Ns. dieneri
LT1n
LT1n.1r
Shangsi
Heller et al. (1988)
Bed
H. parvus
Clarala
P - PZone
T - S Zone
Bed 11
Steiner et al. (1989)
F2
F1
Feixianguan Fm
Daloing Fm
Changhsingian
Hechuan
Ns. triangularis
Ns. homeri
Jialingjiang Fm
Feixianguan Fm
H. parvus
J3
J2
J1
F4
F3
F2
F1
Meishan
Clarala
H. typicalis
H. parvus
LM
P5?
CS
NC
Meishan Mb
Changxing Fm
Yingkeng Fm
Griesbachian
Changhsingian
Bed 15
Bed 25
Hiatus ?,?
South China
300 m
250 m
200 m
150 m
100 m
50 m
0 m
Condensed or hiatus?
Shale
LT1n.1r ?
Ng. changxingensis
Ns. dieneri
Ns. waageni
Ns. gr. bransoni
Ns. crassatus
Ns. discretus; Ns. conservativus
Ns. cristagalli; Ns. pakistanensis
H. parvus
C carb -low
T. ex gr. homeri
Ni. germanica
Ni. kockeli
Ng. regalis
Ns. triangularis
Cs. timorensis
Anisian
Spathian
Smithian
Dienerian
Griesbachian
Lower and Upper Guandao
Talus breccia - no sampling

Figure 10. Latest Permian to Lower Anisian biomagnetostratigraphies from South China (Heller et al., 1988; Steiner et al., 1989; Li and Wang, 1989; Shu and Utting, 1990; Lai et al., 1996; Yin et al., 2001; Nicoll et al., 2002; Lehrmann et al. 2006; Orchard et al. 2007), Iran (Gallet et al., 2000), Italy (Perri and Spalletta, 1998; Scholger et al., 2000; Perri and Farabegoli, 2003), Albania (Muttoni et al., 1996, 2000), and Romania (Grădinaru et al., 2007). South China Conodont zones: Cs—*Neogodolella* (*Clarkina*) *subcarinata*; Nc—*Ng. changxingensis yini*—*Ng. changxingensis*; lm—*Hindeodus latidentatus*–*Ng. meishanensis*; Ns—*Neospathodus*; Cs—*Chiosella*; Ni—*Nicoraella*; T—*Triassospathodus*; Ng—*Neogondolella*. Shangsi ammonoid zones: T-S—Tapashanites-Shevyrevites Assemblage Zone; P-P—Pseudotirolites-Pleuronodoceras Assemblage Zone. The base Anisian used is the first occurrence of *Cs. timorensis* in the Dęsli Caira section (Grădinaru et al., 2007). The composite ammonoid-based biomagnetostratigraphy is slightly modified from the Boreal composite in Figure 9, with the Changhsingian magnetostratigraphy from Steiner (2006), and the radiometric ages from Galfetti et al. (2007b), Lehrmann et al. (2006), and Mundil et al. (2004).

Griesbach Creek (GC2n) is the equivalent of the ~90-m–thick normal magnetozone in the mid parts of the Confederation Point Mb at Creek of Embry (CE3n, Fig. 9). This new correlation resolves the apparently missing magnetozone (i.e., CE2n) at Griesbach Creek using the Ogg and Steiner (1991) correlation and does not violate the regional age assignments used for these formations indicated by Embry (1986).

The association of the $Svalis_1$ palynological assemblage with the Commune Zone in the Svalis Dome area of the Barents Sea indicates the transition from the Boreale Zone to Commune Zone occurs either in the upper part of magnetozone Vh3n or lowermost part of Vh3r. At Griesbach Creek this transition is tightly constrained to be within the lower part of GC1r (i.e., LT1r in the composite of Fig. 9).

The Vh3r magnetozone contains associated late Griesbachian fauna both in Spitsbergen and its correlative magnetozone at Griesbach Creek (GC1r, Fig. 9). The Griesbachian-Dienerian boundary probably occurs within the upper part of Vh3r or Vh4n; it is not tightly constrained by the *Koninckites* sp. fauna at ~57 m or the *B. rosenkrantzi* fauna from the ~38 m level (found in nearby Lusitaniadalen). This is consistent with the Griesbachian-Dienerian boundary placement relative to the magnetostratigraphy in the Abadeh section (Fig. 10; Gallet et al., 2000).

The uppermost part of the Confederation Point Mb in the Griesbach Creek section contains the mid Dienerian ammonoid *V. sverdrupi*, the index species for the second subzone of the Sverdrupi Zone (Tozer, 1994). Hence, magnetozone Vh5n (composite magnetozone LT3n and equivalents CE3n and GC3n) is probably equivalent to most of the lowest part of the Smith Creek Mb at Griesbach Creek and the magnetozone CE3n at Creek of Embry (Fig. 9). In the Vikinghøgda Fm, magnetozone Vh5n appears to be reduced in relative thickness compared to the equivalent (i.e., CE3n and GC3n) magnetozones in the Sverdrup Basin, probably a result of condensation in the shale above the transgressive surface at the base of the Lusitaniadalen Mb. The Boreal Dienerian-Smithian boundary (i.e., base of *H. hedenstroemi* Zone) is probably located in magnetozone LT3n (Vh5n) or in the lower part of LT3r (Vh5r), constrained mostly by the Griesbach Creek data. The revised magnetostratigraphic timescale in the Induan resolves the presence of an additional normal magnetozone in the late Griesbachian proposed by Ogg and Steiner (1991) but questioned by Gallet et al. (2000).

These correlations are supported by the near synchronicity of the transgressive surfaces in magnetozone LT1r at Vikinghøgda, Creek of Embry, and Griesbach Creek, and that within the uppermost part of LT2n at Vikinghøgda and Griesbach Creek (Fig. 9).

Olenekian

Magnetozone Vh6n appears to be subdivided by a reverse magnetozone, which is supported by data from the Creek of Embry and Smith Creek sections (Fig. 9). If the lower normal submagnetozone Vh6n.1n (equivalent to LT4n.1n) is magnetozone GC4n at Griesbach Creek, the base of the Romunderi Zone (indicated by *E. cirratus* at Griesbach Creek) is probably located within Vh6n (or just above). This is compatible with the lowest occurrence of *Arctoceras blomstrandi* in the Vikinghøgda Fm within the base of Vh6r, which is commonly associated with Romunderi Zone fauna (Tozer, 1994).

Correlation between the Vikinghøgda Fm and the Sverdrup Basin sections is not tightly constrained by biostratigraphy around the Smithian-Spathian boundary. *Arctoprionites nodosus* in the upper part of the Smith Creek Mb at Smith Creek is indicative of the Tardus Zone (Tozer, 1994), equating to the upper ~3 m of the Lusitaniadalen Mb and basal part of the Vendomdalen Mb at Vikinghøgda (Fig. 9). The equivalence of the transgressive surface of the Vendomdalen Mb and the Svartfjeld Fm seems to provide the best tie point at this level, which is supported by a variety of lithostratigraphic and sequence stratigraphic data (Mørk et al., 1989, 1992).

The Vendomdalen Mb, which is mostly Spathian, possesses more normal polarity than that of the Svartfjeld Mb in the Creek of Embry section, which is dominated by reverse polarity (Fig. 9). One reason for this may be the truncation of the top of the late Spathian in the Svartfjeld Mb, since the overlying strata are mid Anisian in age (Fig. 9; Ogg and Steiner, 1991). In addition, Lower Spathian ammonoids are not known from the Sverdrup Basin (Dagys and Weitschat, 1993; Tozer, 1994), the ammonoid faunas start in the late Spathian, suggesting possible absence (or condensation) of Lower Spathian strata (Fig. 1). In contrast, we have evidence of Lower Spathian strata from Spitsbergen, an interval which is consistently dominated by normal polarity in both the Vikinghøgda and Milne Edwardsfjellet sections (Fig. 9). The Lower Spathian normal-polarity interval is punctuated by duplicated evidence of a reverse magnetozone (Vh8n.1r and MF1n.2r; LT6r in composite; Fig. 9). We suggest that the magnetostratigraphy of the Svartfjeld Mb is a fuller representation of the early parts of the late Spathian, since it is most similar in polarity character to the mid parts of the Vendomdalen Mb at Milne Edwardsfjellet (Fig. 9). The youngest age from the topmost Svartfjeld Mb at Creek of Embry is questionable, since there is only a single biostratigraphic tie point, when evidence from Milne Edwardsfjellet suggests the Subrobustus Zone has a complex magnetostratigraphic pattern. Similarly, the absence of the Lower Anisian from overlying strata at Creek of Embry indicates the possibility of the erosional removal of part of the uppermost Spathian. Our solution is that magnetozone CE8r probably corresponds to MF1r at Milne Edwardsfjellet (equivalent to LT7r in composite, Fig. 9), and that CE9r is the probable equivalent of Vh8n.2r and MF2n.2r (in the composite LT8r, Fig. 9). The oldest reverse magnetozone in the Svartfjeld Mb (CE7r and SC3r) is probably the equivalent of Vh8n.1r and MF1n.2r (in the composite LT6r), in the transition interval between the early and late Spathian.

THE BIOMAGNETOSTRATIGRAPHY OF THE PERMIAN-TRIASSIC TRANSITION

The Permian-Triassic boundary (PTB) is located at the FAD of the conodont *Hindeodus*

parvus, in the global stratotype section and point (GSSP) at Meishan (Yin et al., 2001). This FAD follows two earlier key events (i.e., in latest Changhsingian)—firstly the extinction event, which is followed a little later by the low in $\delta^{13}C_{carb}$ (Yin et al., 2001; Mundil et al., 2004). At all other locations the PTB placement is based on correlation to the Meishan GSSP (i.e., using conodont, carbon isotopic, sequence stratigraphic, palynological, magnetostratigraphic, and other data). The correlation situation is particularly problematic when using magnetostratigraphy at the PTB since several, poorly documented, restudies of the Meishan sections have given rise to alternative interpretations (Li and Wang, 1989; Zhu and Lui, 1999; Yin et al., 2001; Kozur and Bachmann, 2005). This is compounded by the condensed nature of the GSSP and the poor repeatability and/or intercorrelation between the magnetostratigraphic studies of the latest Permian in South China (Gallet et al., 2000; Steiner, 2006). Two proposals for the position of the PTB in relation to the magnetostratigraphy have been suggested: (a) located within the lower part of a normal magnetozone (Greis-N of Steiner, 2006; LT1n of composite; Fig. 10), which starts in the latest Changhsingian and ends in the mid Griesbachian (Scholger et al., 2000; Yin et al. 2001; Szurlies et al., 2003; Steiner, 2006); and (b) located at the very base of Gries-N/LT1n (Jin et al. 2000b; Nawrocki, 2004). We make the case below that the PTB is located prior to a short-duration reverse polarity interval within the lower part of Gries-N/LT1n.

Tethyan Permian-Triassic Boundary Sections

Both Heller et al. (1988) and Steiner et al. (1989) detect a substantive reverse magnetozone within the lower part of the Feixianguan Fm at Shangsi (in beds 32 and 33; Fig. 10). Here the best marker for the PTB is the FAD of *H. parvus,* which is located in bed 30, in the underlying normal magnetozone (Nicoll et al., 2002; Fig. 10). In the same formation, at Hechuan, a reverse magnetozone occurs ~3 m above the formation base (Steiner et al., 1989), although there its relation to any biostratigraphic datums is not clear (Fig. 10). Consequently at Shangsi and Hechuan, both with substantive sedimentation rates, there is clear evidence of a reverse magnetozone (i.e., LT1n.1r; Figs. 9 and 10), which at Shangsi is younger than the FAD of *H. parvus*. This reverse magnetozone may also be present in the Guandao section (Fig. 10). However, there are also remagnetization problems at Shangsi (and other South China sections, Gallet et al., 2000; Steiner, 2006), in that the several studies that have examined the magnetostratigraphy at Shangsi have proposed slightly different magnetostratigraphies, particularly for the late Permian (Heller et al., 1988; Steiner et al., 1989; Glen et al., 2002).

In the Meishan and Abadeh sections, which have substantially lower sedimentation rates than those at Shangsi and Hechuan, the evidence for the LT1n.1r reverse magnetozone above the PTB is apparently absent (Fig. 10). The interpretation of a reverse magnetozone (in Meishan bed 27) spanning the PTB at Meishan has not been reported in detail (Yin et al., 2001), and apparently has not been validated by further sampling (see personal commun. in Kozur and Bachmann, 2005). The highly condensed nature of the Meishan PTB beds is also clear (Jin et al., 2000a), with some conodont-based evidence suggesting possible hiatus near the boundary (Nicoll et al., 2002). The base of the Abadeh section is also complicated by condensation or omission (Gallet et al., 2000). Hence, the condensed nature of these two sections may explain the lack of evidence for magnetozone LT1n.1r postdating the FAD of *H. parvus*.

At Shangsi the extinction horizon is at the boundary of the Dalong and Feixianguan formations (Wignall et al., 1995), with the low in the $\delta^{13}C_{carb}$ values slightly above this level (~1–5 m; see discussion in Mundil et al., 2004). Such a delay between the extinction horizon and the drop in $\delta^{13}C$ is compatible with microfloral evidence from Greenland (Looy et al., 2001). At Meishan the $\delta^{13}C_{carb}$ low and the marine extinction event are within 10 cm of each other (Jin et al., 2000a; Yin et al., 2001). Recent detailed restudy of the Shangi section places the base of magnetozone LT1n 50 cm above the base of the Feixianguan Fm (Glen et al., 2002), hence just following the extinction event and slightly earlier than the low in $\delta^{13}C_{carb}$.

Italian Dolomites

The PTB interval in the Italian Dolomites is expressed by the transition from the Bellerophon Fm to the Werfen Fm. The lower Griesbachian age of the Werfen Formation is clearly shown by the magnetostratigraphy (Scholger et al., 2000) and the presence of a succession of conodonts, including *H. parvus* at 1.3 m above the base of the Werfen Fm (Perri and Farabegoli, 2003). The terrestrial extinction horizon is located in the Bulla Mb (uppermost part of the Bellerophon Fm; Perri and Farabegoli, 2003) ~0.5 m below the top (Cirilli et al., 1998), where there is a loss of typically late Permian palynomorphs (e.g., *Klausipollenites schaubergeri*, *Jugasporites delesaucei*, *Nuskoisporites dullhuntyi*, and *Paravesicaspora splendens*) and the introduction of forms such as *Densoisporites playfordi*, *Densoisporites nejburgii*, *Convolutispora* sp., and *Rewanispora vermiculata*. This same level also shows evidence of massive soil erosion (Sephton et al., 2005). The character of the palynofloral turnover is similar to that between the Scutasporites unicus–Lunatisporites and Lundbladispora obsoleta–Tympanicysta stoschiana Assemblage Zones at Vikinghøgda, and also to that in Greenland at the boundary of the Schuchert Dal and Wordie Creek formations (Balme, 1980; Looy et al., 2001). Studies of $\delta^{13}C_{carb}$ over this transition in the Alps are contradictory with respect to the placement of the position of the initial isotopic decline, either within the upper part of the Bellerophon Fm (Magaritz et al., 1988; Sephton et al., 2005) or within the base of the Werfen Fm (Holser et al., 1989), reaching a $\delta^{13}C$ low at either ~25m or 10–15 m above the base of the Werfen Fm. In both these cases, the lowest $\delta^{13}C_{carb}$ is younger than the first occurrence of *H. parvus*, clearly at odds with data from other marine sections, implying the $\delta^{13}C_{carb}$ signals from these sediments are not responding to the extinction event (Heydari et al., 2001).

The magnetostratigraphy of the Bellerophon Fm–Werfen Fm transition (Scholger et al., 2000) is fragmented by the unconformities bounding both the lower and upper boundaries of the Bulla Mb (Perri and Farabegoli, 2003); therefore, it is unclear that the succession represents an unbroken record through the latest Permian. The conodonts found in some sections in the uppermost part of the Bellerophon Fm (*H. typicalis* and *H. praeparvus*; Perri and Farabegoli, 2003) are not by themselves diagnostic of pre- or post-parvus strata (Orchard and Krystyn, 1998). In contrast to other studies (e.g., Scholger et al., 2000; Szurlies et al., 2003), we consider it likely that the reverse magnetozone that characterizes the base of the Werfen Fm (lower part of the Tesero Mb) is magnetozone LT1n.1r (i.e., in the Griesbachian), rather than that immediately below LT1n, which is in the Changhsingian. Our interpretation is that the basal part of LT1n magnetozone has been removed by erosion at the basal Werfen Fm unconformity, and the palynofloral turnover horizon in the reverse polarity Bulla Mb occurs in the reverse magnetozone that directly underlies LT1n (Fig. 10). Steiner (2006) from a magnetostratigraphic standpoint has also questioned the lateral and vertical continuity of the Tesero Mb.

Locating precisely the PTB in high-latitude successions is difficult because the primary boundary indicator, the conodont *Hindeodus parvus*, is rare in Boreal successions. Based on correlations to the Otto Fiord section (on Ellesmere Island), where *H. parvus* has been found,

Henderson and Baud (1997) suggested the PTB at Griesbach Creek is likely to be between 15–35 m above the base of the Confederation Point Mb, within the upper part of the range of the ammonoid *O. concavum* (Fig. 9). Similarly, Bjerager et al. (2006) have demonstrated the PTB occurs within the range of *O. concavum* in the East Greenland successions.

In the Vardebukta Fm in the Festningen section in western Spitsbergen, the main negative $\delta^{13}C_{org}$ peak related to the latest Permian extinction event is ~7 m above the base of the formation (Wignall et al., 1998), which equates at Meishan D to some 20 cm below the PTB within the latest Changhsingian (Yin et al., 2001). In Meishan and sections in Greenland this negative excursion in $\delta^{13}C$ is closely associated with (but slightly younger than) the initial palynofloral turnover to assemblages that contain mixed Permian and Triassic miospores (Shu and Utting, 1990; Looy et al., 2001). At Vikinghøgda, the equivalent turnover equates to the boundary between the Scutasporites unicus–Lunatisporites and Lundbladisora obsolete–Tympanicysta stoschiana Assemblage Zones (Fig. 3). In addition, *O. boreale* occurs with opherceritids, indicating the upper part of the Boreale Zone above the PTB. The conodont *N. carinata* at Vikinghøgda also suggests an interval probably younger than *H. parvus* (Henderson and Baud, 1997; Orchard and Tozer, 1997; Yin et al., 2001). Together, these biostratigraphic constraints indicate that the PTB in the Deltadalen Mb is between 1.7 m and ~6 m above the base.

In conclusion, we suggest the following sequence of successive events and/or markers associated with the PTB:

(1) Initiation of strong palynofloral turnover, corresponding to the major terrestrial extinction event (in the late Changhsingian). This may be closely timed to marine biotic extinction events described by many others (Jin et al., 2000a; Looy et al., 2001; Sephton et al., 2005).

(2) Base of magnetozone LT1n, within the latest Changhsingian.

(3) Minimum in $\delta^{13}C$ representing the climax of the late Permian extinctions and its effect on Earth systems.

(4) FAD of *H. parvus* in Chinese sections (indicating the base of the Induan), occurring within the range of *O. concavum* in Boreal sections.

(5) Base of LT1n.1r within the *Otoceras boreale* Zone.

A Magnetostratigraphy for the Lower Triassic

The synthesis of the Boreal-based magnetostratigraphy presented here for the Lower Triassic is broadly similar (but differs in detail) to that of Steiner (2006), who tried to integrate both marine and nonmarine studies. Steiner's approach is inclusive, but data from key marine sections in South China possess problems of detailed correlation internally within South China, let alone the added problems of external correlation (Fig. 10). The age assignments of Steiner (2006) are also strongly framed by the intersection correlations and biostratigraphy of Ogg and Steiner (1991), which we have demonstrated, are flawed. Marine to nonmarine correlations in the Lower Triassic have also been evaluated by Szurlies (2007). The approach adopted here is to illustrate the global applicability of the Boreal composite by using only marine sections. The preliminary magnetostratigraphy from the proposed Induan-Olenekian boundary GSSP at Chaohu presented by Tong et al. (2005) is not reliable (Sun et al., 2007). However, a new set of paleomagnetic data from Chaohu shows a closer match to that from the lower part of the Hechuan section (Sun et al., 2007).

The strongest similarities in the magnetostratigraphy across all marine sections are in the lower parts of the Induan, where magnetozone LT1 can be confidently correlated among all sections (Fig. 10). Similarly the Olenekian-Anisian boundary is well characterized by magnetozone LT9r, which can be correlated between sections at Kçira, Dęsli Caira, Guandao, and Milne Edwardsfjellet (Hounslow et al., 2007b). The placement of the base of the Anisian is likely to be within the Dęsli Caira section in Romania, although the exact marker is not decided (Grădinaru et al., 2007; Hounslow et al., 2007a). It is here informally placed at the first occurrence of the conodont *Ch. timorensis* in the Dęsli Caira section. The putative normal magnetozone LT9r.1n (found in the Milne Edwardsfjellet section; Fig. 9) appears to be supported by at least one correlative normal magnetozone at the upper Guandao section (Orchard et al., 2007). Equivalents to magnetozone LT9r also appear to be present in the Hechuan section, if the conodonts *Neospathodus triangularis* and *Ns. homeri* are used to constrain the correlation to the Guandao sections (Hounslow et al., 2007b; Fig. 10). Based on the match of magnetostratigraphic polarity pattern alone, Steiner (2006) suggested this interval (i.e., LT9r) at Hechuan to be early Spathian, which is not supported by the conodont faunas.

The Dienerian to Smithian interval at Guandao is well characterized with conodont faunas and for the most part is dominated by reverse polarity, until at least mid Smithian. This likely correlates to the interval at Hechuan between the mid parts of the Feixanguan Fm and the first (J1) subdivision of the Jialingjiang Fm. This reverse-dominated interval likely corresponds to the LT2r to LT4r interval in the Boreal composite, a correlation similarly suggested by Steiner (2006) for the Hechuan section. The base of the Olenekian is provisionally defined at the first occurrence of the conodont *Ns. waageni* in the M04 section, Mud, Spiti, India (Krystyn et al., 2007), a level which corresponds to the upper-most part of LT2r (Sun et al., 2007; Fig. 10). A consequence of the intersection correlations in China is that a hiatus or highly condensed interval around the Smithian-Spathian boundary appears to be present in the Guandao and Hechuan sections, a feature hinted at by Steiner (2006) for the Hechuan section. This is probably a reflection of the major climatic, biologic, and sea-level changes at the Smithian-Spathian transition (Payne et al., 2004; Galfetti et al., 2007a, 2007b) and their combined effect on the carbonate platform systems in South China.

CONCLUSIONS

The new magnetostratigraphic data from Spitsbergen allow a robust biomagnetostratigraphy for the Lower Triassic to be constructed. The key changes from previous work are the better magnetostratigraphic definition of the Lower Spathian and the uppermost Spathian provided by the better ammonoid control in the Spitsbergen sections. The magnetostratigraphy over the Dienerian-Smithian boundary interval is also defined with greater confidence by an improved intersection correlation of the Sverdrup Basin data of Ogg and Steiner (1991), prompted by new ammonoid constraints. The magnetostratigraphy from the Deltadalen River section, in combination with data primarily from the Shangsi section (South China), indicates that a short-duration, reverse-polarity magnetozone is present shortly after the first occurrence of *H. parvus*, a magnetostratigraphic event which should allow higher resolution global correlation around the Permian-Triassic boundary interval. The relationship of the new ammonoid-based biomagnetostratigraphy to sections in lower paleolatitudes with conodont biostratigraphies is still problematic around the Smithian-Spathian boundary, which may in part reflect the impact of major environmental and biotic changes on sedimentary systems at this important boundary.

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